Beyond Self-Consistency: Self-Extracted Constraints Do Not Improve LLM Logical Reasoning

Anonymous ACL submission

Can symbolic verification via a provably sound solver improve Large Language Model (LLM) reasoning when constraints are self-extracted from the model's own traces? We find no evidence of improvement. We introduce Symbolic Invariant Consensus Aggregation (SICA), which extracts logical constraints from chain-of-thought traces and selects answers via weighted MAX-SAT. Despite Z3 validating 99% of self-extracted constraints, SICA produces answers identical to majority voting on 90.8% of problems (range 87.3–97.5% across six conditions) spanning six models (7B–32B) and three reasoning domains. No condition shows a significant positive effect after FDR control; four of six powered conditions confirm equivalence within ± 2 pp (TOST), though ten conditions remain underpowered for effects below ~ 2 pp. We systematically diagnose the failure as *symbolic confirmation bias*: self-extracted constraints track the model's majority answer (bias ratios 1.66–$4.97\times$ across nine models, 7B to frontier). We identify the structural mechanism—under source-attribution scoring and empirical degeneration, SICA reduces to majority vote (Proposition 1)—and the $M \to T \to A \to C$ Markov chain that bounds self-extracted information (Remark 1), providing independent evidence consistent with the self-correction limitations of Huang et al. (2024). Twenty-seven training-free variants within the constraint-extraction paradigm fail due to a persistent independence-accuracy trade-off. Preliminary cross-architecture verification escapes the bias ($+6.67$ pp on the weakest generator, $p < 0.001$). Three zero-cost diagnostics characterize failure modes across all 60 tested conditions within this paradigm.

1 Introduction

Self-Consistency (SC) (Wang et al., 2023) selects the majority answer across K chain-of-thought traces, a strategy grounded in the Condorcet Jury Theorem: independent voters converge to truth (de Condorcet, 1785; Ladha, 1992). In practice, SC's gains plateau for strong models (Loo et al., 2025), pointing to correlated traces that reduce the effective ensemble size. A natural remedy is to move beyond simple vote-counting: extract logical constraints from the traces and let a provably sound solver adjudicate. If the solver can filter contradictions and enforce logical consistency, it should recover information that raw majority voting discards.

We test this intuition through Symbolic Invariant Consensus Aggregation (SICA), which extracts logical constraints from reasoning traces (prompted as first-order logic; 99% are propositional at the Z3 level) and selects answers via weighted partial MAX-SAT (§3.1; Figure 1). The result is unequivocal: despite Z3 validating nearly all self-extracted constraints, SICA produces answers identical to majority voting on the vast majority of problems. Across six models (7B–32B), three reasoning domains, and multi-seed replication, no condition shows a significant positive effect after FDR control (Table 1). Four conditions confirm equivalence within ± 2 pp; ten remain underpowered for effects below ~ 2 pp.

The failure is not in the solver but in the constraints it operates on. Self-extracted constraints systematically support the model's majority answer, a phenomenon we term *symbolic confirmation bias* (§5.1). Because each trace determines both an answer and its constraints, the formal channel reduces to a weighted echo of the majority vote. We prove this degeneration is structural (Proposition 1) and show it holds across nine models from 7B to frontier scale. Prior work shows LLMs cannot self-correct reasoning without external feedback (Huang et al., 2024); we provide independent evidence that this limitation extends to formal self-verification.

Twenty-seven training-free variants within

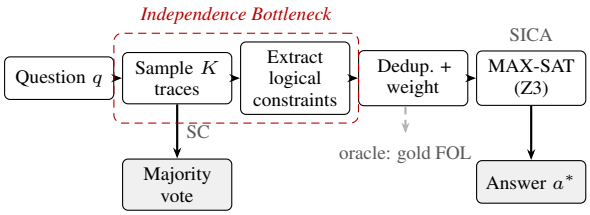


Figure 1: The SICA pipeline. SC selects the majority answer from K traces. SICA extracts logical constraints, deduplicates, and solves MAX-SAT via Z3. Red dashed box: the independence bottleneck. Dashed arrow: oracle variant with gold FOL.

the constraint-extraction paradigm confirm the trade-off is fundamental (§5.2): methods that break trace correlation destroy accuracy, while methods that preserve accuracy fail to achieve independence. Cross-architecture verification provides preliminary evidence of escape: a structurally independent verifier yields significant gains when it exceeds the generator's accuracy, with benefit governed by a capability-gap pattern between verifier and generator (§6). Three zero-cost diagnostics enable practitioners to predict a priori whether any self-extraction-based aggregation method will help (§5.3).

Our contributions are:

1. **The symbolic mirage.** Provably sound Z3 verification becomes vacuous under self-extraction. We systematically diagnose symbolic confirmation bias as the mechanism, identify the structural degeneration (Proposition 1), and confirm it empirically across nine models and 27 training-free variants, localizing the failure to extraction rather than verification (§5.1, §5.2).
2. **Diagnostic framework.** Three zero-cost diagnostics characterize failure modes across all tested conditions within the self-extraction paradigm, with prospective validation on held-out model families, providing practitioners with a priori failure-mode diagnosis from a pilot run (§5.3).
3. **Cross-architecture escape.** Preliminary evidence that structurally independent verification can break the bottleneck, with gain governed by a capability-gap pattern between verifier and generator (§6).

2 Related Work

Self-consistency and ensemble theory. Self-consistency (SC) selects the most frequent answer across chain-of-thought traces (Wang et al., 2023). Extensions address free-form generation (Chen et al., 2024), confidence weighting (Taubenfeld et al., 2025), diverse prompting (Li et al., 2023), and reasoning-aware filtering (Wan et al., 2024; Toh et al., 2024; Kim et al., 2026). SC gains plateau early (Loo et al., 2025; Byerly and Khashabi, 2026). The Condorcet Jury Theorem (de Condorcet, 1785) guarantees majority-vote convergence under independence; Ladha (1992) show correlated voters reduce effective ensemble size. Our K_{eff} adapts Fleiss' κ from ensemble diversity theory (Kuncheva and Whitaker, 2003; Brown et al., 2005) to quantify this compression (§3.1). Kim et al. (2025) demonstrate correlated errors in LLM outputs; Kuai et al. (2026) propose cross-model independence auditing, complementary to our within-model process-κ.

Neuro-symbolic reasoning. We distinguish three paradigms: *specification-generation* (Pan et al., 2023; Olausson et al., 2023; Ye et al., 2024) translates problems into formal specifications before answering; *program-generation* (Gao et al., 2023; Chen et al., 2023b; Lyu et al., 2023) produces executable code; both formalize *before* answer generation. *Trace-extraction* (this work) operates post-hoc, inheriting confirmation bias (§5.1). Recent extensions expand these paradigms with symbolic planning, logic agents, and verified reasoning (Ryu et al., 2025; Xu et al., 2026; Li et al., 2025; Wang et al., 2025; Singh et al., 2026; Luo et al., 2025; Miya, 2025); Amonkar et al. (2025) show formalization quality degrades with problem complexity; Chan et al. (2026) argue logical soundness alone is unreliable for neurosymbolic verification. Prior studies (Olausson et al., 2023; Amonkar et al., 2025) note high formalization error rates but treat them as quality problems; we show the errors are *systematically biased* toward the model's answer. Proposition 1 establishes that self-extracted constraints reduce SICA to majority vote under degeneration ($C \perp Y \mid A$; Figure 3), making self-verification an information bottleneck. Huang et al. (2024) show LLMs cannot self-correct without external feedback; our findings provide independent evidence consistent with this limitation in formal verification.

Distinction from CoT confirmation bias. Symbolic confirmation bias (SCB) is complementary to, but structurally distinct from, CoT-level confirmation bias (Wan et al., 2025; Turpin et al., 2024; Jhaveri et al., 2026). CoT bias operates at the *trace level*—individual chains may not faithfully reflect the model's computation—while SCB operates at

the *aggregation level*: even if each trace is locally faithful, the *collection* of self-extracted constraints mirrors the answer distribution. Methodologically, SCB is formalized via a Markov chain structure (Proposition 1, Remark 1) and confirmed through systematic exclusion of 27 training-free alternatives (§5.2). The key distinction: CoT bias implies individual traces may be unfaithful; SCB shows that even Z3-validated constraints carry no information beyond majority voting when self-extracted.

Test-time compute and self-verification. Scaling test-time compute via additional traces (Snell et al., 2025), structured search (Yao et al., 2023), or extended reasoning (Guo et al., 2025; Qwen Team, 2024) faces the same correlation challenge when traces share a single model's biases. Multi-agent debate (Liang et al., 2023; Du et al., 2024), self-verification and refinement (Weng et al., 2023; Madaan et al., 2023; Dhuliawala et al., 2023), iterative self-improvement (Shinn et al., 2023; Zhou et al., 2024), tool-augmented critique (Gou et al., 2024), formal CoT verification (Zhu et al., 2025; Chen et al., 2025), and multi-agent verification (Lifshitz et al., 2025) report gains, but none tests self-extracted constraints under formal solvers (cf. Kamoi et al., 2024). We find the bottleneck lies in constraint extraction, not verification: self-extracted constraints inherit the model's bias regardless of solver soundness (§5.1). Oracle controls confirm the verification mechanism itself is sound (Appendix G).

Training-based verification. The methods above operate at inference time without additional training. Trained verifiers can escape the self-extraction bottleneck via external reward signals: outcome and process reward models (Cobbe et al., 2021; Lightman et al., 2024; Wang et al., 2024), RL-based self-verification (Zeng et al., 2025), dual preference optimization (Yang et al., 2025), and execution-based filtering in code (Li et al., 2022; Chen et al., 2023a). Proposition 1 addresses inference-time self-extraction; trained verifiers operate outside this pipeline, potentially breaking the information ceiling (Remark 1) by violating the trace-determined assumption.

3 Method

3.1 SICA Pipeline

Self-Consistency (SC) (Wang et al., 2023) samples K reasoning traces and selects the majority answer. Under correlated traces (Fleiss' κ (Fleiss, 1971)), effective sample size reduces to

$$K_{\text{eff}} = K \,/\, [1 + (K-1) \cdot \kappa] \tag{1}$$

(following Ladha (1992)); e.g., K=12 traces with κ=0.69 yield $K_{\text{eff}} \approx 1.3$, meaning twelve correlated traces carry no more independent information than 1.3 uncorrelated ones. This motivates constraint-based aggregation.

Symbolic Invariant Consensus Aggregation (SICA) extends SC by extracting formal constraints from traces and selecting the answer best supported by the satisfiable subset (Figure 1).

Trace generation. Given question q, an LLM generates K traces $\{t_1, \ldots, t_K\}$ at temperature $T = 0.7$. Each trace t_i yields answer a_i.

Constraint extraction. The same LLM extracts logical constraints from each trace at $T_{\text{ext}} = 0.3$. The extraction prompt requests JSON output with constraint type, expression, and Z3-compatible formula (Appendix A); in practice, 99.0% of extracted constraints are propositional at the Z3 level, with only 1.0% using true quantifiers. The *extraction rate* is the fraction of traces yielding at least one parseable constraint.

Deduplication and MAX-SAT. Constraints are deduplicated by syntactic equivalence (global average deduplication ratio: 46.0%); each unique constraint c_m receives weight w_m equal to the number of source traces. The objective finds the largest-weight satisfiable subset:

$$\mathcal{C}^* = \arg\max_{\mathcal{S} \subseteq \mathcal{C}} \sum_{c \in \mathcal{S}} w_c \quad \text{s.t.} \quad \mathcal{S} \text{ is satisfiable.} \tag{2}$$

Z3 (de Moura and Bjørner, 2008) solves this with a 10-second timeout. Answer assertion constraints (`answer_a == True`, weight 1) ensure every candidate receives a nonzero base score, making SICA a strict generalization of SC.

Scoring. Let $\mathcal{C}^{\setminus} = \mathcal{C} \setminus \mathcal{C}^*$. Each candidate a is scored by:

$$\text{score}(a) = \sum_{c \in \mathcal{C}^*} w_c \cdot \sigma(c, a) - \alpha \sum_{c \in \mathcal{C}^{\setminus}} w_c \cdot \sigma(c, a), \tag{3}$$

where $\sigma(c, a) = 1$ if c was extracted from any trace producing answer a, and $\alpha = 0.5$. The selected answer is $a^* = \arg\max_a \text{score}(a)$.

Degeneration condition. If self-extracted constraints are coupled to their source answer, MAX-

SAT retains nearly all ($|\mathcal{C}^{\setminus}| \approx 0$), and scoring degenerates to:

$$\text{score}(a) \approx \sum_{\substack{c \in \mathcal{C}^* \\ \text{source}(c) \to a}} w_c, \qquad (4)$$

equivalent to majority voting. We show in §5.1 that this degeneration occurs in practice.

The C_0 diagnostic. The multi-candidate rate $C_0 = |\{q : |\mathcal{A}(q)| > 1\}|/|\mathcal{Q}|$ is a zero-cost diagnostic: $C_0 < 5\%$ means SICA reduces to SC. $\alpha = 0.5$ is not critical (± 0.49 pp across $\alpha \in [0, 1]$).

Bias ratio. We quantify extraction bias via the *bias ratio*:

$$\text{BR}(M, T) = \frac{P(c \text{ supports } a_M \mid a_M \neq a^*)}{P(c \text{ supports } a^* \mid a_M \neq a^*)}, \quad (5)$$

where c is a self-extracted constraint, a_M the model's majority answer, and a^* the ground truth. BR quantifies the degree to which extracted constraints align with the majority answer; values above 1 are expected under source-attribution scoring (Proposition 1) and reflect both the base-rate effect of trace-count distributions and any additional extraction alignment. BR $= 1$ indicates unbiased extraction. We validate this metric empirically in §5.1.

3.2 Experimental Setup

Models. Qwen2.5-14B-Instruct, Qwen2.5-7B-Instruct, Qwen3-14B (non-thinking mode) (Yang et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), LLaMA-3.1-8B-Instruct (Dubey et al., 2024), and QwQ-32B (Qwen Team, 2024) (7B–32B). We also evaluate the universality of symbolic confirmation bias via cross-model extraction on five frontier models (gpt-4o, gpt-4.1, gpt-5.5, o3, Gemini-2.5-Pro; Table 2); Mistral-7B extracts constraints from target traces since closed-source APIs do not support self-extraction.

Datasets. FOLIO (Han et al., 2022) (n=204, with gold logical annotations), ProofWriter depth-5 (Tafjord et al., 2021) (n=600), LogiQA (Liu et al., 2020) (n=200). StrategyQA (Geva et al., 2021) (n=200) serves as a domain-formalizability diagnostic. GSM8K (Cobbe et al., 2021) (exploratory, n=30) and AIME 2024 (n=22) probe the degenerate regime ($C_0 < 5\%$).

Parameters. K=12 traces at T=0.7, extraction at T_{ext}=0.3, α=0.5, Z3 timeout 10s. McNemar exact test (α=0.05); 95% bootstrap CIs (10,000 resamples) for $n \geq 200$. SICA incurs 4.5× SC wall-clock; extraction accounts for 76% of overhead.

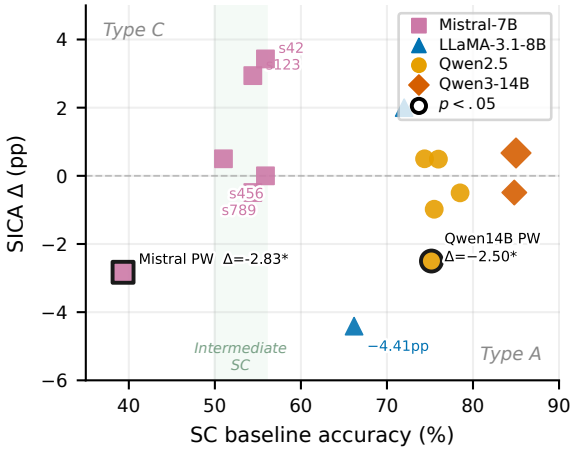


Figure 2: SICA Δ (pp) vs. SC baseline. Filled: $n \geq 100$; open: $n < 100$. The intermediate-SC region (50–60%, shaded) contains the only positive trends (none reaching $p < 0.05$). Type C: nominally significant degradation (raw $p = 0.014$; BH adj. $p = 0.112$).

4 Results

Table 1 reports accuracy across all non-degenerate self-extraction conditions sorted by SC baseline. No condition shows a significant SICA effect after FDR control (BH, m=16). Two conditions show nominally significant degradation: Mistral-7B on ProofWriter (-2.83 pp, raw $p = 0.014$) and Qwen2.5-14B on ProofWriter (-2.50 pp, raw $p = 0.011$); both reach adjusted $p = 0.112$. An aggregate sign test yields 8 positive and 7 negative Δ values ($p = 1.00$, two-sided).

Figure 2 reveals the regime structure underlying these results. Fleiss' κ ranges from 0.21 to 0.82 across conditions, yielding $K_{\text{eff}} = 1.20$–3.62 at $K = 12$ (Table 19, Appendix). Positive Δ requires both sufficient K_{eff} (≤ 1.30 never improves) and reasonable SC accuracy ($K_{\text{eff}} = 3.62$ with SC $= 39\%$ degrades rather than helps). Three regimes emerge: Type C (SC $\leq 40\%$), where low base accuracy causes error amplification; an intermediate band (50–60%), containing the only positive trends; and Type A (SC $\geq 60\%$), where symbolic confirmation bias dominates. Seed-level noise is substantial: across three Mistral-7B seeds, $\kappa \in [0.41, 0.51]$ yet Δ ranges from $+2.94$ to -2.45 pp, indicating that constraint quality varies beyond what independence alone predicts. We trace the failure mechanism in §5.

TOST equivalence tests (Lakens, 2017) with margin ± 2 pp—adopted as effects below one per-

Table 1: SICA vs. SC across all non-degenerate self-extraction conditions (sorted by SC). No condition shows a significant gain after BH correction (m=16). Δ = SICA − SC (pp). Symbols: ^{b}gold ICL; **bold**: $p < 0.05$; * uncorrected $p < 0.05$ (BH adj. $p = 0.112$); ^{3s} 3-seed; ^{4s} 4-seed confirmatory (unmarked: single-seed exploratory). All conditions have McNemar power $< 80\%$ for $|\Delta| = 2$ pp (Appendix D). Degenerate conditions ($C_0 < 5\%$: QwQ-32B AIME n=22, Qwen2.5-14B GSM8K n=30; exploratory) yield $\Delta = 0$; oracle upper bounds in Appendix G.

Model	Domain	n	SC (%)	SICA (%)	Δ (pp)	p
Type C: error amplification (SC ≤ 40%)						
Mistral-7B	PW-D5	600	39.33	36.50	**−2.83**	0.014*
Intermediate SC (SC ∼50–60%, Mistral-7B only)						
Mistral-7B	LogiQA	200	50.00	51.50	+1.50	0.581
Mistral-7B	FOLIO (s123)	204	54.41	57.35	+2.94	0.070
Mistral-7B^{3s}	FOLIO	204	56.53	56.70	+0.16	1.000
Mistral-7B	FOLIO (s456)	204	55.88	55.88	0.00	1.000
Mistral-7B	FOLIO (s789)	204	59.31	56.86	−2.45	0.267
Type A: confirmation bias (SC ≥ 60%)						
LLaMA-3.1-8B	FOLIO	204	66.18	61.76	−4.41	0.093
LLaMA-3.1-8B	StrategyQA	200	72.00	74.00	+2.00	0.424
Qwen2.5-14B	FOLIO (ICL)^b	199	74.37	74.87	+0.50	1.000
Qwen2.5-14B	PW-D5	600	74.00	71.50	**−2.50**	0.011*
Qwen2.5-14B^{4s}	FOLIO	204	75.01	73.78	−1.24	1.000
Qwen2.5-7B	FOLIO	204	75.98	76.47	+0.49	1.000
Qwen2.5-14B	LogiQA	200	79.00	78.00	−1.00	0.625
Qwen3-14B	FOLIO	204	83.33	84.80	+1.47	1.000
Qwen3-14B	PW-D5	600	85.83	85.67	−0.17	1.000
Qwen3-14B (think)	FOLIO	204	86.27	86.76	+0.49	1.000

centage point per class provide negligible practical benefit on datasets with ≥3 answer categories—confirm four conditions as equivalent; ten remain underpowered for effects below ∼2 pp (MDE 1.86–4.84 pp; Appendix D). TOST sensitivity analysis across three margins (±1, ±2, ±3 pp) confirms that equivalence conclusions are margin-dependent (Appendix U). Absence of evidence is not evidence of absence for small effects. Three-seed replication on Mistral-7B confirms seed-level noise (mean $\Delta = +0.16 \pm 2.70$ pp; Table 33). Four-seed replication on Qwen2.5-14B × FOLIO confirms the pattern ($\Delta \in [0.00, -2.01]$ pp).

Oracle controls replacing self-extracted constraints with gold annotations confirm the MAX-SAT mechanism is sound (+29.90/+13.73 pp FOLIO, +17.17 pp ProofWriter; $p < 0.001$), yet self-extraction captures only ∼12% of this ceiling (Appendix G).

5 The Symbolic Mirage

5.1 Symbolic Confirmation Bias

Self-extracted constraints systematically align with the model's majority answer. On Mistral-7B FOLIO, when SICA errs, 83.3% of constraints support the incorrect answer—bias ratio $4.97\times$ (Table 16)—and 94% of errors overlap with SC. Z3 rejects only 3.4% of constraints, and MAX-SAT produces answers identical to count-only aggregation on 94.4% of problems (Table 10), confirming that the formal verification layer is vacuous (Eq. 4): 99% of constraints pass Z3 validation yet add no information beyond majority voting. The α-insensitivity confirms degeneration: under $|\mathcal{C}^{\setminus}| \approx 0$ the penalty term vanishes and α becomes irrelevant, producing ≥99% agreement across $\alpha \in \{0.1, \ldots, 0.9\}$ as a ceiling effect (Table 11).

Empirically, confirmation bias extends beyond incorrect instances. On problems where SC selects the correct answer, constraints still disproportionately support the majority ($\text{BR}_{\text{correct}}$ 1.29–$3.64\times$ across all six tested conditions, with 5/6 showing $\text{BR}_{\text{correct}} \geq \text{BR}_{\text{wrong}}$), confirming that self-extracted constraints track distributional frequency rather than logical validity. A permutation null model (1000 permutations, $p > 0.7$) confirms BR is driven by trace-count distributions, not constraint content. Base-rate decomposition shows SICA scoring compresses rather than amplifies this effect: observed BR falls consistently below base-rate BR across all six conditions (mean excess -1.85), as deduplication and MAX-SAT attenuate the majority-answer advantage. The bias is structural: it is inherent to the majority-vote geometry (Proposition 1), not an artifact of extraction.

The pattern generalizes: $4.97\times$ (Mistral), $3.85\times$ (Qwen-14B), $2.88\times$ (LLaMA), all > 1.0 with 95% bootstrap CIs (Table 16). Cross-model

Table 2: Symbolic confirmation bias across nine models on FOLIO ($n = 204$). BR: mean per-problem bias ratio. n_{err}: number of incorrect instances on which BR is computed. Δ: SICA − SC (pp). Ext.: [a]self-extraction; [b]cross-model (Mistral-7B extracts FOL from target traces). *Interim ($n = 95/204$). All models exhibit BR > 1; BR estimates for high-accuracy models ($n_{\text{err}} < 40$) have wider confidence intervals; Gemini ($n_{\text{err}} \approx 12$) should be interpreted with particular caution. Process-level similarity (κ_{proc}) ranges from 0.25 (R1-Distill) to 0.79 (gpt-4o/4.1/Gemini); see Table 21.

Model	Ext.	SC (%)	n_{err}	BR	Δ (pp)
Mistral-7B	[a]	55.88	90	4.97	0.00
LLaMA-8B	[a]	—	—	2.88	—
Qwen-14B	[a]	75.98	49	3.85	−0.49
R1-Distill-8B	[b]	74.51	52	2.92	+2.45
gpt-4o	[a]	73.91	53	2.94	−1.80
gpt-4.1	[b]	82.84	35	2.70	−1.47
gpt-5.5	[b]	84.31	32	3.48	+0.49
o3	[b]	82.65	35	1.66	0.00
Gemini-2.5-Pro*	[b]	87.37	12	2.79	−1.10

extraction confirms universality across nine models from 7B to frontier (BR 1.66–4.97×; Table 2). BR > 1 further holds across all eight cells of a 2×2 generator×extractor matrix on FOLIO and ProofWriter (range 1.94–2.70; Appendix W). Constraints use trace-specific variables (56.4% zero cross-answer overlap), collapsing aggregation to majority voting.

Two controls rule out answer leakage. BR is uniform across early, mid, and late reasoning steps (Fisher $p = 0.71$); high-consensus errors amplify to BR $= 18.8\times$ versus $2.70\times$ for low-consensus ($p < 0.001$). Removing the answer sentence from traces before extraction yields no BR decrease (Wilcoxon $p \geq 0.12$) despite significant accuracy loss (McNemar $p < 0.013$). Symbolic confirmation bias reflects a distributional prior embedded throughout reasoning, not answer contamination.

Reasoning-process overlap exceeds answer-level correlation: on Mistral-7B FOLIO, mean process similarity (pairwise TF-IDF cosine; Appendix O) reaches 0.69, far exceeding answer-level κ (0.41–0.51 across seeds). The gap generalizes across four families (Table 21). Under cross-model extraction, SICA gain decreases monotonically with process similarity ($\rho = -1.0$, $p = 0.008$; Table 17). Conditional correction analysis confirms the same pattern: cross-architecture NLI yields positive correction ratios while same-model SICA yields negative ones, with correction ratio correlating with combo gain ($\rho = 0.84$, $p = 0.014$; Appendix C).

These patterns—universal BR > 1, structural equivalence to trace-count distributions, and process-level correlation exceeding answer-level agreement—follow from a structural property of self-extraction pipelines.

Proposition 1 *(Argmax Preservation).* Let $t_1, \ldots, t_K \overset{\text{i.i.d.}}{\sim} P_M(\cdot \mid q)$ with trace-determined answers $a_i = h(t_i)$ and $n_a = |\{i : a_i = a\}|$. Under (i) the degeneration condition $|\mathcal{C}^{\setminus}| \approx 0$ (Eq. 4), (ii) source-attribution scoring where $\sigma(c, a)=1$ iff c originates from a trace producing a (Eq. 3), and (iii) approximately uniform constraint yield per trace within each answer group, SICA reduces to majority vote: $\arg\max_a \text{score}(a) = \arg\max_a n_a$.

Proof. By (i), $\mathcal{C}^* \approx \mathcal{C}$ and the penalty term vanishes: $\text{score}(a) = \sum_{c:\,\sigma(c,a)=1} w_c$. By (ii), each constraint contributes only to its source answer. By (iii), total weight for answer a is proportional to n_a, preserving the argmax. Empirically, SICA's answer selection agrees with majority vote on 90.8% of problems across six conditions (range 87.3–97.5%; Table 10). □

Remark 1 *(Information Ceiling).* If constraints are trace-determined ($c_i = f(t_i)$), then $Y — T_{1:K} — C_{1:K}$ forms a Markov chain, and the data processing inequality gives $I(C_{1:K}; Y) \leq I(T_{1:K}; Y)$: self-extracted constraints cannot carry more information about Y than the source traces. This bound holds for any model, task, and deterministic f, applying to any self-extraction pipeline. The contribution is the identification of the $M \to T \to A \to C$ Markov chain structure, establishing that constraints carry no information about Y beyond what A already provides.

Remark 2. Proposition 1 assumes degeneration and source attribution. Under *semantic* evaluation, the equivalence additionally requires *answer alignment*: $P(c_i \text{ supports } a_i) \geq P(c_i \text{ supports } a')$ for $a' \neq a_i$, which is structurally motivated and empirically validated (BR $\in [1.66, 4.97]$; Table 2), but could fail if f systematically extracted premises contradicting their source trace's conclusion. Since Proposition 1 shows SICA reduces to majority vote under degeneration, BR ≥ 1 follows from the trace-count geometry; the empirical BR consistently exceeds this floor (1.66×–4.97×; Table 2), reflecting both base-rate effects and extraction alignment.

Since the bias holds for any model, task, and extraction function f, training-free remediation strategies face a structural limitation: methods that

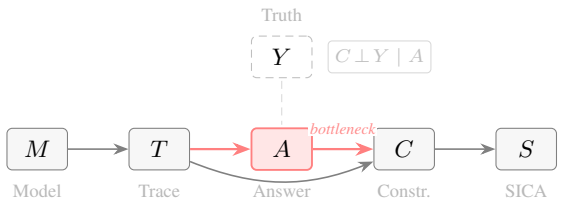


Figure 3: Causal DAG of the self-extraction pipeline. All information about ground truth Y reaching constraints C is mediated by answer A; Remark 1 formalizes this as an information ceiling via the data processing inequality.

preserve extraction accuracy necessarily inherit the model's distributional prior.

5.2 Why Remediation Fails

All 27 training-free variants within the constraint-extraction paradigm, spanning four categories, produce neutral-to-negative results (Figure 4; Appendices K–M). The pattern reveals a fundamental independence-accuracy trade-off: methods that successfully break trace correlation simultaneously destroy accuracy, while methods that preserve accuracy fail to achieve independence.

Four representative failure modes illustrate this trade-off. *Debate* ($\kappa = 0.15$, $\Delta = -18.6$ pp) breaks correlation but collapses accuracy. *Cross-model extraction* decreases κ slightly but $\Delta \approx 0$, because same-family models share systematic biases (Appendix Table 18). *Contrastive prompting* produces imbalanced output (94 pro vs. 65 con), indicating confirmation bias transfers to constraint generation. *Iterative refinement* reduces surface duplication but refined constraints still align with the majority answer.

All non-debate strategies cluster at $\kappa \approx 0.40$–0.47, a same-model ceiling across three families: regardless of prompting strategy, self-extraction cannot produce constraints independent of the model's distributional prior. The oracle ceiling ($+29.90$ pp with gold constraints) proves the mechanism is sound; self-extraction captures only $\sim$12%. No individual strategy achieves significance after correction (Appendix X.1), suggesting symbolic confirmation bias is intrinsic to self-referential constraint generation within the extraction-aggregation paradigm.

5.3 Diagnostic Framework

Three zero-cost diagnostics predict the outcome of any self-extraction-based aggregation method from a $K = 12$ pilot run: (1) the degenerate-regime indicator $C_0 < 5\%$ (SICA reduces to SC); (2)

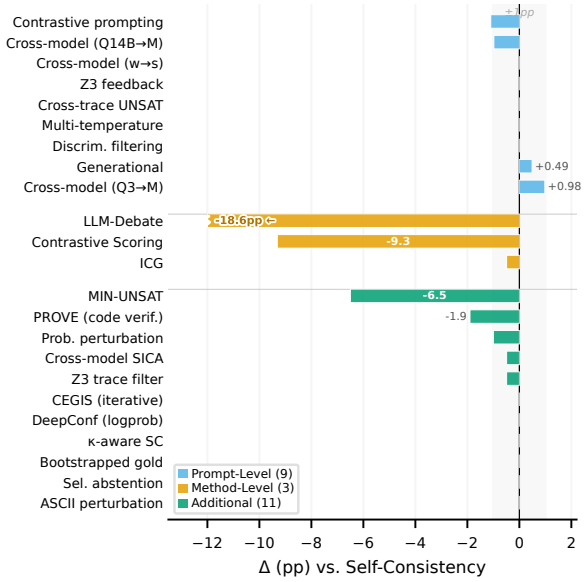


Figure 4: Twenty-three of 27 training-free strategies across four categories (four additional strategies tested but omitted for clarity). All produce neutral-to-negative Δ vs. SC. Grey band: ± 1 pp noise zone.

the diversity floor $K_{\text{eff}} \leq 1.30$ (insufficient vote diversity for any aggregation gain); (3) the bias ratio BR combined with SC regime (Type A: SC $\geq 60\%$ with BR > 1 predicts confirmation-bias degradation; Type C: SC $\leq 40\%$ predicts error amplification). These boundaries correctly characterize the failure mode across all 60 tested conditions (16 main configurations, 27 remediation variants on Mistral-7B$\times$FOLIO, and 17 analytical conditions) within the self-extraction paradigm. The framework's practical value lies in failure-mode diagnosis—degenerate, diversity-limited, or confirmation-biased—from a K=12 pilot run, enabling practitioners to rule out self-extraction before full deployment. Conversely, when diagnostics detect sufficient capability gap (§6), the framework correctly predicts positive gain ($+6.67$ pp for Mistral-7B on ProofWriter).

Decision boundaries ($C_0 < 5\%$, $K_{\text{eff}} \leq 1.30$, SC $< 40\%$) were derived from Mistral-7B and Qwen2.5-14B. We evaluate prospectively on two held-out generators: LLaMA-3.1-8B (different family) on FOLIO, where the framework predicts Type A bias and the observed $\Delta = -4.41$ pp confirms; and Qwen3-14B (same family, different generation) on FOLIO and ProofWriter, likewise confirmed ($\Delta \leq +1.47$ pp, n.s.).

6 Escaping the Bottleneck

Motivation. No training-free self-extraction method achieves independence from the model's

distributional prior (§5.2). Remark 1 implies that any escape must break the self-extraction dependency: the information ceiling $I(C;Y) \leq I(T;Y)$ binds whenever constraints are deterministic functions of the generating model's traces. We test whether a structurally independent verifier, sharing neither architecture nor training data with the generator, provides corrective signal that escapes this bound.

Method. We use DeBERTa-large (He et al., 2021) (406M, encoder-only), fine-tuned on MultiNLI (Williams et al., 2018), to classify (premise, conclusion) pairs as entailment, contradiction, or neutral. In combo mode, each NLI-supporting vote counts w times alongside SC's K votes. We evaluate on Mistral-7B FOLIO ($n = 204$) and ProofWriter depth-5 ($n = 600$).

Zero-shot results. DeBERTa NLI alone achieves 49.51% on FOLIO and 55.00% on ProofWriter. Combo w=3 yields +6.67 pp on ProofWriter for Mistral-7B ($p < 0.001$, significant after BH correction) and +2.94 pp on FOLIO (n.s.); all generators with SC $\geq$ 72% show no significant improvement ($\Delta \leq +0.83$ pp). Same-model SICA *degrades* ProofWriter (-2.83 pp; Table 1), illustrating that corrective signal must originate outside the model's distribution.

Capability gap. Four generators spanning a 47-point SC range on ProofWriter (n=600), all paired with DeBERTa-large-mnli (55.00% standalone; Table 3): only Mistral-7B (SC 39.33%, below the verifier) gains significantly (+6.67 pp, $p < 0.001$); all three above the verifier show no significant improvement ($\Delta \leq +0.83$ pp). These results point to a *capability-gap pattern*: verification gain decreases monotonically with generator accuracy, reaching breakeven near the verifier's standalone accuracy. The relationship holds across three NLI architectures and six verifier-generator pairs (Spearman $\rho = 1.0$, $p = 0.003$, n=6; Figure 8), suggesting a structural property of independent verification rather than an architecture artifact. Per-label error complementarity diminishes as the generator improves (Appendix V).

Domain-adapted verifier. Fine-tuning DeBERTa on ProofWriter depth-5 OWA (99.67% standalone) amplifies the escape: combo gains reach +12.67 pp (Mistral-7B) and +11.67 pp (LLaMA-8B), a 14× amplification over zero-shot, with near-total error complementarity (Table 29, Appendix). Domain-adapted verification extends escape to generators with SC up to ∼72%, consistent with the prediction that the capability gap can be shifted by increasing verifier accuracy. A cost-aware cascade recovers 83–90% of full-combo gain at ∼35% NLI cost (Appendix U).

Table 3: Cross-generator NLI escape with DeBERTa-large-mnli verifier (55.00% standalone, ProofWriter n=600). Combo gain decreases monotonically with generator capability. Only Mistral-7B (SC below the verifier) shows a significant gain.

Generator	Family	SC	Gap	Combo Δ (w=3)	p
Mistral-7B	Mistral	39.33%	+15.67	+6.67 pp	<0.001
LLaMA-3.1-8B	LLaMA	72.50%	−17.50	+0.83 pp	0.665
Qwen2.5-14B	Qwen	74.00%	−19.00	+0.67 pp	0.503
Qwen3-14B	Qwen	85.83%	−30.83	+0.50 pp	0.45

7 Discussion and Conclusion

Our central finding is that the value of formal verification depends on the *independence* of its inputs, not the soundness of its logic. Three diagnostics (K_{eff}, BR, C_0) operationalize this principle and correctly characterize failure modes across all 60 tested conditions within the self-extraction paradigm, with decision boundaries generalizing prospectively to held-out model families (§5.3). The theoretical bounds (Proposition 1, Remark 1) apply to any deterministic self-extraction pipeline by construction; empirical validation covers 27 training-free variants within the SICA paradigm.

This independence principle has direct consequences for test-time compute scaling (Snell et al., 2025): with $K_{\text{eff}} \leq 3.62$ for K=12, same-model traces scale redundancy, not information. The principle applies to any self-verification method that draws on the generating model's own distribution, providing independent evidence consistent with Huang et al. (2024). Escaping the bottleneck requires structural independence, capability ordering, and task-verifier alignment (§6); whether trained extractors can shift this boundary remains open. The diagnostics we provide enable future work to recognize this limitation upfront.

Limitations

Open-source models range from 7B to 32B; frontier models confirm SCB universality via cross-model extraction. All datasets are English-language logical reasoning benchmarks; we do not evaluate trained extractors or multilingual settings. Statistical power is limited for effects below ∼2 pp; the capability-gap analysis rests on n=6 conditions (Appendix Y).

References

Rikhil Amonkar, Ceyhun Efe Kayan, Qimei Lai, Ronan Le Bras, and Li Zhang. 2025. A reality check of

language models as formalizers on constraint satisfaction problems. *arXiv preprint arXiv:2505.13252*.

Gavin Brown, Jeremy Wyatt, Rachel Harris, and Xin Yao. 2005. Diversity creation methods: A survey and categorisation. *Information Fusion*, 6(1):5–20.

Nicholas Byerly and Daniel Khashabi. 2026. Self-consistency falls short! analyzing the effectiveness and limitations of self-consistency in mathematical reasoning. *arXiv preprint*.

Jason Chan, Robert Gaizauskas, and Zhixue Zhao. 2026. Position: Logical soundness is not a reliable criterion for neurosymbolic fact-checking with LLMs. *arXiv preprint arXiv:2604.04177*.

Bei Chen, Fengji Zhang, Anh Nguyen, Daoguang Zan, Zeqi Lin, Jian-Guang Lou, and Weizhu Chen. 2023a. CodeT: Code generation with generated tests. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*.

Jiefeng Chen, Jie Ren, Xinyun Chen, Chengrun Yang, Ruoxi Sun, Jinsung Yoon, and Sercan Ö. Arık. 2025. SETS: Leveraging self-verification and self-correction for improved test-time scaling. *arXiv preprint arXiv:2501.19306*.

Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. 2023b. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *Transactions on Machine Learning Research*.

Xinyun Chen, Renat Aksitov, Uri Alon, Jie Ren, Kefan Xiao, Pengcheng Yin, Sushant Prakash, Charles Sutton, Xuezhi Wang, and Denny Zhou. 2024. Universal self-consistency for large language model generation. In *ICML*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Marie Jean Antoine Nicolas de Caritat de Condorcet. 1785. *Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix*. Imprimerie Royale, Paris.

Leonardo de Moura and Nikolaj Bjørner. 2008. Z3: An efficient SMT solver. In *TACAS*.

Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2023. Chain-of-verification reduces hallucination in large language models. *arXiv preprint arXiv:2309.11495*.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2024. Improving factuality and reasoning in language models through multiagent debate. In *ICML*.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, and 1 others. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Joseph L. Fleiss. 1971. Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382.

Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.

Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. 2021. Did Aristotle use a laptop? a question answering benchmark with implicit reasoning strategies. *TACL*.

Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Minlie Huang, Nan Duan, and Weizhu Chen. 2024. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *ICLR*.

Daya Guo, Dejian Yang, He Zhang, Junxiao Song, and 1 others. 2025. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Simeng Han, Hailey Schoelkopf, Yilun Zhao, Zhenting Qi, Martin Riddell, Luke Benson, Lucy Sun, Ekaterina Zubova, Yujie Qiao, Matthew Burtell, David Peng, Jonathan Fan, Yi Liu, Brian Wong, Malcolm Sailor, Ansong Ni, Linyong Nan, Jungo Kasai, Tao Yu, and 7 others. 2022. FOLIO: Natural language reasoning with first-order logic. *arXiv preprint arXiv:2209.00840*.

Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2021. DeBERTa: Decoding-enhanced BERT with disentangled attention. In *Proceedings of ICLR*.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large language models cannot self-correct reasoning yet. In *ICLR*.

Ayush Rajesh Jhaveri, Anthony GX-Chen, Ilia Sucholutsky, and Eunsol Choi. 2026. Failing to falsify: Evaluating and mitigating confirmation bias in language models. *arXiv preprint arXiv:2604.02485*.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.

Ryo Kamoi, Yusen Kobayashi, Tharindu Goyal, Rui Hou, Hanjie Sun, and James Caverlee. 2024. When can LLMs actually correct their own mistakes? a

critical survey of self-correction of LLMs. *Transactions of the Association for Computational Linguistics*, 12:1–20.

Elliot Kim, Avi Garg, Kenny Peng, and Nikhil Garg. 2025. Correlated errors in large language models. *arXiv preprint arXiv:2506.07962*.

Junseok Kim, Nakyeong Yang, Kyungmin Min, and Kyomin Jung. 2026. Reliability-aware adaptive self-consistency for efficient sampling in LLM reasoning. *arXiv preprint arXiv:2601.02970*.

Chenchen Kuai, Jiwan Jiang, Zihao Zhu, Hao Wang, Keshu Wu, Zihao Li, Yunlong Zhang, Chenxi Liu, Zhengzhong Tu, Zhiwen Fan, and Yang Zhou. 2026. How independent are large language models? A statistical framework for auditing behavioral entanglement and reweighting verifier ensembles. *arXiv preprint arXiv:2604.07650*.

Ludmila I. Kuncheva and Christopher J. Whitaker. 2003. Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51(2):181–207.

Krishna K. Ladha. 1992. The Condorcet jury theorem, free speech, and correlated votes. *American Journal of Political Science*, 36(3):617–634.

Daniël Lakens. 2017. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362.

Jundong Li and 1 others. 2025. Aristotle: Mastering logical reasoning with a logic-complete decompose-search-resolve framework. In *ACL*.

Yifei Li, Zeqi Lin, Shizhuo Zhang, Qiang Fu, Bei Chen, Jian-Guang Lou, and Weizhu Chen. 2023. Making language models better reasoners with step-aware verifier. In *ACL*.

Yujia Li, David Choi, Junyoung Chung, Nate Kushman, Julian Schrittwieser, and 1 others. 2022. Competition-level code generation with AlphaCode. In *Science*, volume 378, pages 1092–1097.

Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Zhaopeng Tu, and Shuming Shi. 2023. Encouraging divergent thinking in large language models through multi-agent debate. *arXiv preprint arXiv:2305.19118*.

Shalev Lifshitz, Sheila A. McIlraith, and Simon S. Du. 2025. Multi-agent verification: Scaling test-time compute with multiple verifiers. *arXiv preprint arXiv:2502.20379*.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let's verify step by step. In *ICLR*.

Jian Liu, Leyang Cui, Hanmeng Liu, Dandan Huang, Yile Wang, and Yue Zhang. 2020. LogiQA: A challenge dataset for machine reading comprehension with logical reasoning. In *IJCAI*, pages 3622–3628.

Ryan Loo and 1 others. 2025. Reevaluating self-consistency scaling in multi-agent systems. *arXiv preprint arXiv:2511.00751*.

Yiqi Luo and 1 others. 2025. From ambiguity to verdict: A semiotic-grounded multi-perspective agent for LLM logical reasoning. *arXiv preprint arXiv:2509.24765*.

Qing Lyu, Shreya Havaldar, Adam Stein, Li Zhang, Delip Rao, Eric Wong, Marianna Apidianaki, and Chris Callison-Burch. 2023. Faithful chain-of-thought reasoning. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, and 1 others. 2023. Self-refine: Iterative refinement with self-feedback. In *NeurIPS*.

Shinobu Miya. 2025. Eidoku: A neuro-symbolic verification gate for LLM reasoning via structural constraint satisfaction. *arXiv preprint arXiv:2512.20664*.

Theo X. Olausson, Alex Gu, Benjamin Lipkin, Cedegao E. Zhang, Armando Solar-Lezama, Joshua B. Tenenbaum, and Roger Levy. 2023. LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers. In *EMNLP*.

Liangming Pan, Alon Alber, Wenhu Xu, Yixin Deng, Huan Chen, and Heng Ji. 2023. Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning. In *EMNLP Findings*.

Qwen Team. 2024. QwQ: Reflect deeply on the boundaries of the unknown. Technical Blog. https://qwenlm.github.io/blog/qwq-32b-preview/.

Hyun Ryu, Gyeongman Kim, Hyemin S. Lee, and Eunho Yang. 2025. Divide and translate: Compositional first-order logic translation and verification for complex logical reasoning. In *International Conference on Learning Representations (ICLR)*.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Vikash Singh, Darion Cassel, Nathaniel Weir, Nick Feng, and Sam Bayless. 2026. VERGE: Formal refinement and guidance engine for verifiable LLM reasoning. *arXiv preprint arXiv:2601.20055*.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. 2025. Scaling LLM test-time compute optimally can be more effective than scaling parameters. In *ICLR (Oral)*.

Oyvind Tafjord, Bhavana Dalvi, and Peter Clark. 2021. ProofWriter: Generating implications, proofs, and abductive statements over natural language. In *ACL Findings*.

Asaf Taubenfeld, Ido Goldstein, and Oren Tsur. 2025. Confidence improves self-consistency in LLMs. In *ACL Findings*.

Vernon Y.H. Toh, Deepanway Ghosal, and Soujanya Poria. 2024. Not all votes count! programs as verifiers improve self-consistency of language models for math reasoning. *arXiv preprint arXiv:2410.12608*.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2024. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *NeurIPS*.

Guangya Wan, Yuqi Wu, Jie Chen, and Sheng Li. 2024. Reasoning aware self-consistency: Leveraging reasoning paths for efficient LLM sampling. *arXiv preprint arXiv:2408.17017*.

Yue Wan, Xiaowei Jia, and Xiang Lorraine Li. 2025. Unveiling confirmation bias in chain-of-thought reasoning. In *Findings of the Association for Computational Linguistics: ACL 2025*.

Peiyi Wang, Lei Li, Zhihong Shao, R.X. Xu, Damai Dai, Yifei Li, Deli Chen, Y. Wu, and Zhifang Sui. 2024. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *ACL*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *ICLR*.

Yilun Wang and 1 others. 2025. FoVer: First-order logic verification for natural language reasoning. *TACL*.

Yixuan Weng, Minjun Zhu, Fei Xia, Bin Li, Shizhu He, Shengping Liu, Bin Sun, Kang Liu, and Jun Zhao. 2023. Large language models are better reasoners with self-verification. In *Findings of EMNLP*.

Adina Williams, Nikita Nangia, and Samuel Bowman. 2018. A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of NAACL-HLT*, pages 1112–1122.

Lei Xu, Pierre Beckmann, Marco Valentino, and Andre Freitas. 2026. Adaptive LLM-symbolic reasoning via dynamic logical solver composition (DyLoS). In *EACL*.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Wang, and 1 others. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Chenxu Yang and 1 others. 2025. DuPO: Dual process optimization for scalable self-correction of LLMs. *arXiv preprint arXiv:2508.14460*.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36.

Xi Ye, Qiaochu Chen, Isil Dillig, and Greg Durrett. 2024. Satlm: Satisfiability-aided language models using declarative prompting. In *Advances in Neural Information Processing Systems*, volume 36.

Ruotian Zeng, Yuanzhi Liu, and 1 others. 2025. S^2R: Teaching LLMs to self-verify and self-correct via reinforcement learning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. 2024. Language agent tree search unifies reasoning acting and planning in language models. In *International Conference on Machine Learning (ICML)*.

Hao Zhu, Yuxuan Gao, Nuo Xu Ke, and Pan Lu. 2025. Vericot: Verified chain of thought. *arXiv preprint arXiv:2511.04662*.

A Prompt Templates

SICA uses domain-specific prompt templates for constraint extraction: one for mathematical reasoning (GSM8K, AIME) and one for logical reasoning (ProofWriter, FOLIO). Both share the same system prompt. The LLM is called with temperature T=0.3, max tokens 1024, and structured JSON output mode. Up to 6 traces are processed in parallel via thread pooling, with 2 retries on failure (exponential backoff, base 0.5s).

System Prompt.

```
You are a constraint extraction tool.
    Output ONLY
valid JSON. Do NOT solve problems,
    reason about
correctness, or think step-by-step.
    Extract
constraints from the given trace
    exactly as stated.
```

Mathematical Constraint Extraction (EXTRACTION_PROMPT). Used for GSM8K and AIME datasets. The placeholder `{trace}` is replaced with the reasoning trace at runtime.

```
Extract mathematical constraints from
    this reasoning
trace as a JSON object.

Rules:
```

```
- type: "equation", "inequality", or
    "predicate"
- expression: human-readable, e.g. "x
    + y == 14"
- z3_formula: a single comparison
    expression in
  Python Z3 syntax
- source_step: step number where the
    constraint
  appears
- Use variable names from the trace

z3_formula MUST follow these rules:
- Allowed operators: ==, !=, <, >,
    <=, >=, +, -, *,
  /, **, %
- Use individual named variables: a1,
    a2, a3...
  NOT arrays a[1] or sets A
- Expand all sums explicitly:
  "a1 + a2 + a3 == 55" NOT "sum([a1,
     a2,a3])"
- NO Python builtins: sum(), len(),
    max(), min(),
  list(), set(), range(), int()
- Each z3_formula must be exactly ONE
    comparison
  (==, !=, <, >, <=, >=)
- For multiple values, write separate
    constraints
  for each

Output ONLY the JSON below, nothing
    else:
{"constraints": [{"type": "equation",
  "expression": "x + y == 14",
  "z3_formula": "x + y == 14",
  "source_step": 1}],
 "answer": "final_answer",
 "variables": ["x", "y"]}

Trace:
{trace}
```

Logical Constraint Extraction (LOGIC_EXTRACTION_PROMPT). Used for ProofWriter and FOLIO datasets.

```
IMPORTANT: You are an EXTRACTION tool
    , NOT a problem
solver.
- Do NOT re-solve or re-reason about
    the problem.
- Do NOT verify whether the trace's
    reasoning is
  correct.
- ONLY extract the logical
    constraints that are
  explicitly stated or used in the
     trace below.
- Output ONLY the JSON object,
    nothing else.

Extract logical constraints from this
     reasoning trace
about a logic problem.

The problem has facts (base
    assertions), rules
(grounded implications), and derived
    conclusions.
Convert each into a Z3 boolean
    formula.

Variable naming:
- Unary property: property_entity
  (e.g., kind_anne, red_charlie,
     round_fiona)
- Binary relation:
    relation_subject_object
  (e.g., eats_bear_squirrel,
     likes_lion_cat)
- Use lowercase with underscores only

Constraint types:
- "fact": a given assertion,
  e.g., kind_anne == True
- "rule": a grounded implication over
     specific
  entities, e.g., Implies(kind_anne,
     nice_anne)
- "derived": an inferred conclusion
    from facts +
  rules, e.g., nice_anne == True

Z3 operators:
- Implies(a, b) for if-then rules
- And(a, b, ...) for conjunctive
    conditions
- Or(a, b, ...) for disjunctive
    conditions
- Not(a) for negation
- == True / == False for boolean
    assertions

Output ONLY the JSON below, nothing
    else:
{"constraints": [{"type": "fact",
  "expression": "kind(anne)",
  "z3_formula": "kind_anne == True",
  "source_step": 1}],
 "answer": "True or False or Unknown
     ",
 "variables": ["kind_anne"]}

Trace:
{trace}
```

Output Schema. Both prompts produce a JSON object with the following structure:

```
{
  "constraints": [
    {
      "type": "<equation|inequality|
          predicate>
                or <fact|rule|derived
                    >",
      "expression": "<human-readable
          description>",
      "z3_formula": "<Z3-compatible
          expression>",
      "source_step": <integer>
    }
  ],
  "answer": "<final answer string>",
  "variables": ["<var1>", "<var2>",
      "..."]
```

```
}
```

The pipeline additionally injects an `answer_assertion` constraint per trace (not produced by the LLM):

```
{"type": "answer_assertion",
 "expression": "answer is <ans>",
 "z3_formula": "__sica_answer_<ans>
     == True",
 "source_step": -1}
```

Prompt Selection Logic. Domain assignment is determined by dataset: FOLIO and ProofWriter use LOGIC_EXTRACTION_PROMPT; GSM8K, AIME, StrategyQA, and ARC-Challenge use EXTRACTION_PROMPT. StrategyQA and ARC-Challenge are assigned to the mathematical prompt because their reasoning structure does not involve formal logic predicates or implications, despite StrategyQA producing boolean answers.

B Z3 MaxSAT Evaluation Pseudocode

Algorithm 1 describes the Z3 MaxSAT solving procedure. Algorithm 2 describes the source-trace attribution scoring.

Domain constraint variables (e.g., `x + y == 14`) and answer assertion variables (e.g., `__sica_answer_42 == True`) are logically independent in the Z3 formulation. Discriminative signal arises solely from conflicts among domain constraints: when traces supporting different answers produce mutually contradictory constraints, the solver excludes the lower-weight subset, and the scorer penalizes the corresponding answers.

Algorithm 1 Z3 Weighted Partial MAX-SAT

Require: Deduplicated constraint set $\mathcal{C} = \{(c_m, w_m, S_m)\}_{m=1}^{M}$ where c_m is a Z3 formula, w_m is weight, S_m is the set of source trace indices

Ensure: Partition $(\mathcal{C}^*, \mathcal{C}^{\setminus})$ where $\mathcal{C}^*$ is the max-weight satisfiable subset

opt $\leftarrow$ Z3.OPTIMIZE()
for $m = 1, \ldots, M$ **do**
  $b_m \leftarrow$ fresh Boolean indicator
  opt.ADD($b_m \Rightarrow c_m$) ▷ soft clause guarded by indicator
  opt.ADDSOFT(b_m, weight $= w_m$)
end for
result $\leftarrow$ *opt*.CHECK()
model $\leftarrow$ *opt*.MODEL()
$\mathcal{C}^* \leftarrow \{(c_m, w_m, S_m) : \mathit{model}(b_m) = \top\}$
$\mathcal{C}^{\setminus} \leftarrow \mathcal{C} \setminus \mathcal{C}^*$
return $(\mathcal{C}^*, \mathcal{C}^{\setminus})$

Algorithm 2 Source-Trace Attribution Scoring

Require: Satisfied set $\mathcal{C}^*$, excluded set $\mathcal{C}^{\setminus}$, candidate answers $\mathcal{A}$, trace-answer mapping τ : trace_idx $\to a$, penalty weight α

Ensure: Score $s(a)$ for each $a \in \mathcal{A}$

for each $a \in \mathcal{A}$ **do**
  $T_a \leftarrow \{i : \tau(i) = a\}$ ▷ trace indices producing answer a
  $\mathit{pos}(a) \leftarrow \sum_{(c,w,S)\in\mathcal{C}^*} w \cdot \mathbf{1}[S \cap T_a \neq \emptyset]$
  $\mathit{neg}(a) \leftarrow \sum_{(c,w,S)\in\mathcal{C}^{\setminus}} w \cdot \mathbf{1}[S \cap T_a \neq \emptyset]$
  $s(a) \leftarrow \mathit{pos}(a) - \alpha \cdot \mathit{neg}(a)$
end for
return $a^* = \arg\max_{a\in\mathcal{A}} s(a)$

C Capability-Accuracy Constraint Diagnostics

Conditional Correction Ratio. We define the *Conditional Correction Ratio* (CCR) to summarize verifier–generator complementarity:

$$\mathrm{CCR}(V, G) = \log \frac{P(V \text{ correct} \mid G \text{ wrong})}{P(V \text{ correct} \mid G \text{ correct})}. \tag{6}$$

CCR > 0 means V preferentially corrects G's errors; CCR < 0 indicates shared bias. CCR links our findings: (1) SCB corresponds to $\mathrm{CCR}(\text{self}, G) \approx 0$; (2) Finding 2 predicts $\mathrm{CCR}(V, G) > 0$ when V is independent and capable.

Empirical CCR across verifier–generator pairs. Table 4 reports CCR for eight verifier–generator–dataset conditions, sorted by CCR descending. All three cross-architecture NLI pairs on ProofWriter yield CCR > 0 (verifier preferentially corrects generator errors), while all three same-model SICA conditions yield CCR < 0 (shared bias). CCR correlates with combo gain: Spearman $\rho = 0.839$, 95% bootstrap CI $[0.338, 0.990]$; LOO-robust (min $p = 0.014$). Four of eight conditions share the DeBERTa verifier, yielding effective $n < 10$; results should be interpreted as a consistent pattern rather than a precise effect size.

Per-condition diagnostics. Table 5 reports per-condition diagnostic metrics referenced in §M.

D Statistical Power Analysis

We compute the minimum detectable effect (MDE) for each condition at $\alpha = 0.05$ and 80% power using exact binomial power calculations over Mc-

Table 4: Empirical CCR across eight verifier–generator conditions (sorted by CCR descending). CCR > 0: cross-architecture NLI verifiers on ProofWriter. CCR < 0: same-model SICA or mismatched capability gap.

Generator	Verifier	Dataset	CCR	Δ_{maj} (pp)
Mistral-7B	BART-MNLI	PW-600	+1.86	+5.17
Mistral-7B	DeBERTa-MNLI	PW-600	+1.67	+6.67
Mistral-7B	RoBERTa-MNLI	PW-600	+1.42	+4.17
Qwen2.5-14B	DeBERTa-MNLI	PW-600	−0.21	+0.33
Mistral-7B	DeBERTa-MNLI	FOLIO-204	−0.47	+2.94
Mistral-7B	SICA	FOLIO-204	−2.93	+3.43
Qwen2.5-14B	SICA	PW-600	−4.21	−4.50
Mistral-7B	SICA	PW-600	−4.61	−2.83

Model	Domain	C_0 (%)	Extr. Rate	Contr. Rate	Δ (pp)
Qwen2.5-14B	FOLIO	35.3	0.89	0.022	0.00
Qwen2.5-14B	ProofWriter	39.2	0.89	0.022	+0.33
Qwen2.5-14B	StrategyQA†	10.0	0.70	—	0.00
Qwen2.5-14B	ARC-Chall.†	5.0	1.00	0.067	+2.00
Qwen2.5-7B	FOLIO+PW†	42.1	0.94	0.027	0.00
Mistral-7B	FOLIO+PW†	50.0	0.88	0.056	0.00
LLaMA-3.1-8B	StrategyQA	41.0	—	—	+2.00
LLaMA-3.1-8B	FOLIO	61.8	0.94	0.062	−4.41
DeepSeek-R1-7B	FOLIO+PW†	58.3	0.84	0.037	−3.33
QwQ-32B	Combined†	2.6	—	—	0.00
Qwen2.5-14B	GSM8K†	13.3	0.15	—	0.00

Table 5: Capability-accuracy constraint diagnostics. C_0: fraction of questions with multiple distinct candidate answers across $K = 12$ traces. Extraction rate: fraction of traces yielding at least one parseable logical constraint. Contradiction rate: fraction of extracted constraint sets with internal contradictions. Δ: SICA−SC gap (pp). †: $n \leq 60$. Dashes indicate conditions where extraction diagnostics were not computed (StrategyQA uses boolean answers without symbolic extraction; QwQ-32B Combined had near-zero C_0, making extraction analysis uninformative). This table includes additional exploratory conditions not reported in Table 1.

Nemar discordant pairs. Table 6 reports three grouped conditions with $n \geq 200$ (Qwen2.5-14B FOLIO and ProofWriter are pooled into a single n=804 analysis, corresponding to four Table 1 rows), which constitute our confirmatory analyses. The remaining seven conditions ($n \leq 60$) yield infinite MDE due to insufficient discordant pairs and serve only as regime characterization.

The largest study (Qwen-14B, n=804, 30 discordant pairs) can detect effects as small as 1.86pp, yet the observed Δ of +0.25pp is an order of magnitude smaller. The two LLaMA-8B conditions require effects of 4.64pp and 4.84pp to reach significance; neither observed Δ approaches these thresholds. All three conditions are consistent with the null hypothesis that SICA provides no accuracy gain over self-consistency.

Condition	n	MDE	Obs. Δ	Obs. Pow.
Qwen-14B FOLIO+PW	804	1.86pp	+0.25pp	5.7%
LLaMA-8B StrategyQA	200	4.64pp	+2.00pp	20.6%
LLaMA-8B FOLIO	204	4.84pp	−4.41pp	18.6%

Table 6: Power analysis for confirmatory conditions ($n \geq 200$). MDE = minimum detectable effect at 80% power. All observed effects fall below their respective MDEs, and observed power remains below 21% in every case.

E Bootstrap Confidence Intervals

We compute 95% confidence intervals for Δ(SICA−SC) via 10,000 bootstrap resamples (percentile method, seed 42). Table 7 reports the results for the three confirmatory conditions.

Condition	n	Obs. Δ	95% CI
Qwen-14B FOLIO+PW	804	+0.25pp	[−1.12, +1.62]
LLaMA-8B StrategyQA	200	+2.00pp	[−1.50, +5.50]
LLaMA-8B FOLIO	204	−4.41pp	[−5.88, +0.98]

Table 7: Bootstrap 95% confidence intervals for Δ(SICA − SC) in percentage points. All intervals include zero, consistent with no reliable accuracy difference.

All three intervals contain zero. The narrowest interval (Qwen-14B, [−1.12, +1.62]) rules out practically meaningful gains (>2pp) even in the most favorable case. The widest interval (LLaMA-8B FOLIO, [−5.88, +0.98]) skews negative, indicating that SICA may slightly hurt accuracy in this condition. These results corroborate the McNemar tests and power analysis: no condition provides evidence that logical constraint aggregation improves over majority voting.

F Case Studies: Discordant Predictions

Among 804 problems, SICA and SC agree on 773 (96.1%). The 31 discordant cases comprise 16 SICA wins (SICA correct, SC wrong), 14 SC wins (SC correct, SICA wrong), and 1 where both select different wrong answers. All 31 occur at exact ties (entropy=1.0, 6:6 vote split) or near-ties (entropy≥0.9183, 8:4 at most), since SICA can differ from SC only when constraint weights shift the majority away from SC's tiebreak choice. We present three representative cases that illustrate the mechanism behind capability-accuracy constraint failures.

Case 1: SICA Wins (folio_134, FOLIO). *Premises*: Some mammals have teeth. Platypus have no teeth. Platypus are mammals. Humans have teeth. *Conclusion*: Humans are mammals. *Gold*: Unknown.

The premises state that *some* mammals have teeth and that humans have teeth, but "having teeth" is not sufficient for "being a mammal" under open-world semantics. SC produces a 6:6 split (Unknown:6, True:6) and tiebreaks alphabetically to True (wrong). SICA extracts 7 constraints; MaxSAT excludes 1 constraint inconsistent with the rest, reducing the weighted score for True-voting traces and tipping the aggregation to Unknown (correct). The excluded constraint likely captured an incorrect inference step (e.g., affirming the consequent) present only in True-voting traces. While SICA succeeds here, the success is coincidental: the excluded constraint happens to anti-correlate with the correct answer.

Case 2: SC Wins (ProofWriter-1066_Q6, ProofWriter). *Key chain*: White(Bob)→Round(Bob) [white things are round] → Young(Bob) [all round things are young]. *Conclusion*: "Bob is not young." *Gold*: False (Bob IS young).

SC produces a 6:6 split (False:6, True:6) and tiebreaks to False (correct). SICA extracts 17 constraints; MaxSAT excludes 2 that appear preferentially in False-voting traces. These excluded constraints likely captured correct intermediate derivations (e.g., `young_bob == True`) that appeared only in traces completing the full deduction chain. By excluding them, MaxSAT removes evidence supporting the correct answer. SICA selects True (wrong). This case illustrates how self-extracted constraints can systematically mislead: correct reasoning steps become discriminative signals that the solver mistakenly penalizes.

Case 3: Non-discriminative Constraints (ProofWriter-903_Q6, ProofWriter). *Key chain*: Nice(Mouse)→Cold(Mouse) [all nice people are cold] → Chases(Mouse,Mouse) [nice and cold → chase mouse]. *Conclusion*: "The mouse does not chase the mouse." *Gold*: False.

SC produces a 6:6 split (False:6, Unknown:6) and tiebreaks to False (correct). SICA extracts 44 constraints but excludes only 1 (2.3%). The remaining 43 are domain facts and grounded rules (`nice_mouse == True`, `Implies(nice_mouse, cold_mouse)`, etc.) that Z3 satisfies identically regardless of the final answer. Both False-voting and Unknown-voting traces share the same premises and grounded rules; they differ only in whether the trace completed the full derivation chain. The single excluded constraint carries negligible weight against 43 shared constraints, so SICA collapses to SC.

Pattern. These cases exemplify the general pattern across the 773 agreeing problems: the vast majority of extracted constraints encode shared domain structure rather than answer-specific reasoning, and MaxSAT aggregation degenerates to majority vote. Table 8 summarizes.

Table 8: Summary of three representative discordant cases. All occur at exact 6:6 vote ties. Constraint exclusion can break ties in either direction (Cases 1–2) or have no effect (Case 3).

Case	ID	Gold	SC	SICA	#Cons
1 (SICA win)	folio_134	Unknown	True ×	Unknown ✓	7
2 (SC win)	PW-1066_Q6	False	False ✓	True ×	17
3 (Non-disc.)	PW-903_Q6	False	False ✓	False ✓	44

G Oracle Control

Replacing self-extracted constraints with gold annotations (FOLIO logical formulae, ProofWriter DSL), Oracle-SICA yields $+29.90$ pp on Mistral-7B FOLIO ($p < 10^{-13}$), $+13.73$ pp on Qwen2.5-14B FOLIO ($p = 2.0 \times 10^{-7}$), and $+17.17$ pp on ProofWriter ($p < 10^{-14}$). Z3 agrees with gold labels on 95.1% (FOLIO) and 97.1% (ProofWriter). The MAX-SAT mechanism is sound; Mistral-7B's best self-extracted Δ ($+3.43$ pp) captures only 11.5% of its oracle ceiling. For Qwen2.5-14B on ProofWriter, the gap is wider still: self-extraction degrades accuracy (-2.50 pp) against an oracle ceiling of $+17.17$ pp, indicating that stronger models do not self-extract more effectively.

ProofWriter per-subtype analysis. ProofWriter problems encode deductive rules as Domain-Specific Language (DSL) logic programs. Each problem uses one of four predicate subtypes: attribute predicates with or without negation (AttNeg, AttNoneg), and relation predicates with or without negation (RelNeg, RelNoneg). Relation predicates involve binary relations between entities (e.g., `eats(bear, squirrel)`), while attribute predicates assign unary properties (e.g., `kind(anne)`).

The oracle pipeline parses these DSL annotations deterministically into Z3 formulas without any LLM involvement. Of the 600 ProofWriter problems, 594 (99.0%) produce valid Z3 parses; the remaining 6 fail due to unsupported DSL constructs. Across all 7,128 trace-problem pairs sub-

mitted to Z3, the solver returns a result in every case (0% solver errors). Z3 evaluations agree with ground-truth labels on 577 of 594 parseable problems (97.14%), confirming that the deterministic DSL-to-Z3 translation faithfully preserves problem semantics.

Table 9 reports the per-subtype breakdown. Relation-predicate subtypes show substantially larger gains than attribute-predicate subtypes: RelNeg improves by $+17.86$ pp and RelNoneg by $+13.01$ pp, compared to $+9.09$ pp for AttNoneg and $+6.87$ pp for AttNeg. Relation predicates involve longer reasoning chains with more intermediate entities, producing greater answer diversity across traces (C_0) and more opportunities for oracle constraints to correct errors.

Table 9: ProofWriter oracle experiment per-subtype breakdown. Oracle-SICA uses gold DSL annotations parsed deterministically to Z3 for consistency checking. Relation-predicate subtypes (RelNeg, RelNoneg) show larger gains than attribute-predicate subtypes; relation subtypes involve longer reasoning chains and benefit more from oracle constraints. Bold indicates best accuracy per row.

Subtype	n	SC Acc (%)	Oracle Acc (%)	Δ^a (pp)
AttNoneg	154	87.01	**98.70**	+11.69
AttNeg	160	85.00	**91.90**	+6.90
RelNoneg	146	66.44	**84.20**	+17.76
RelNeg	140	60.00	**89.30**	+29.30
Overall	600	74.00	**91.17**	+17.17

^{a} n=582 (extraction failures excluded).

H Constraint Ablation Details

Table 10: Constraint ablation (Qwen2.5-14B, $n = 804$). Agreement: predictions identical to SC. MAX-SAT and count-only produce nearly identical predictions, confirming that self-extracted constraints carry no signal beyond vote counts.

Aggregation	Acc. (%)	Agreement w/ SC
SC (majority vote)	76.24	—
SICA (MAX-SAT)	76.49	773 / 804
Count-only	76.62	774 / 804
Random-weight	75.87	764 / 804

I Remediation Strategy Details

J Additional Intervention Results

K Extraction Error Taxonomy

Constraints must discriminate between candidate answers for MAX-SAT to produce a ranking different from majority vote. We measure discriminative power at two levels on Qwen2.5-14B across FOLIO and ProofWriter ($n = 804$). At the source-trace level, 39.2% of total constraint weight originates from traces whose answer differs from the plurality, but only 12.1% of individual constraints fall into this minority category. At the Z3 semantic level, 91.8% of constraints [95% CI: 91.2, 92.4] are technically discriminative, producing different satisfiability outcomes when evaluated against different candidate answers. Only 8.2% are non-discriminative.

The high discriminative rate does not translate into accuracy gains because discrimination mirrors the model's own answer distribution. Each trace produces constraints with its own variable names, so Z3 treats identically-intended propositions as unrelated symbols; among cross-answer constraint pairs, 56.4% share zero Z3 variables. This variable isolation means constraints from different traces are trivially jointly satisfiable, and MAX-SAT retains nearly all of them regardless of source answer. Removing the 8.2% non-discriminative constraints yields zero accuracy improvement. The discrimina-

Table 11: MAX-SAT vs. count-only agreement across constraint weight α on FOLIO ($n = 204$). Mistral-7B (BR $= 4.97\times$, seed 42); Qwen2.5-14B (BR $= 3.85\times$, seed 123). Agreement $\geq 99\%$ at all α (reflecting ceiling effect as the main SICA effect is near zero), confirming vacuity is weight-insensitive.

α	Mistral Agr. (%)	Qwen2.5 Agr. (%)
0.1	100	100
0.3	100	100
0.5	100	100
0.7	99	100
0.9	99	100

Table 12: Nine prompt-level remediation strategies. Baseline: Qwen2.5-14B FOLIO, SC $= 75.98\%$, $\Delta = -0.49$ pp. None reaches significance. Three method-level strategies are reported in Table 13.

Strategy	Target	Δ (pp)	Key finding
Contrastive^a	Prompt	-1.10	non-disc. 6.0% (↓ from 8.2%)
Cross-model^b	Architecture	0.00	disc. 0% (weak→strong); ±1
Qwen3-14B^c	Capability	-0.49	$p = 0.655$
Z3 feedback	Solver	—	0.9% UNSAT trigger
Cons. filtering	Noise	0.00	$T{\geq}4 \rightarrow$ SC
Cross-trace UNSAT	Independence	—	56.4% $\emptyset$ vars
T=1.0^d	Diversity	0.00	isolated peak at T=0.7 only
Multi-temp^e	Diversity	0.00	b=5, c=5 (balanced)
Disc. filtering^f	Filtering	0.00	803/804 agree w/ SICA

^{a} Mistral-7B extracts from Qwen2.5-14B traces. ^{b} Strong→weak: Qwen3-14B (60.29%) and Qwen2.5-14B (58.33%) extract from Mistral-7B FOLIO traces (n=204); self-extraction baseline 59.31%. ^{c} n=204, SC $= 83.33\%$. Mistral-7B FOLIO, n=204. ^{e} Mistral-7B FOLIO, $T \in \{0.3, 0.7, 1.2\}$, 4 traces each, n=204. ^{f} Qwen2.5-14B, n=804; removing non-discriminative constraints.

Table 13: Method-level interventions on Mistral-7B with McNemar's paired test. All produce neutral-to-catastrophic results. ICG shows two FOLIO seeds; p for the first. *** $p < 0.001$; ** $p < 0.01$.

		FOLIO ($n = 204$)		LogiQA ($n = 200$)	
Method	Target	Δ (pp)	p	Δ (pp)	p
SC baseline	—	0.00	—		
ICG	Trace coupling	−1.96 / +1.47	.206		
Contrastive	Scoring bias	**−9.31**	.007		
Debate	Role diversity	**−18.63**	<.001	**−23**	

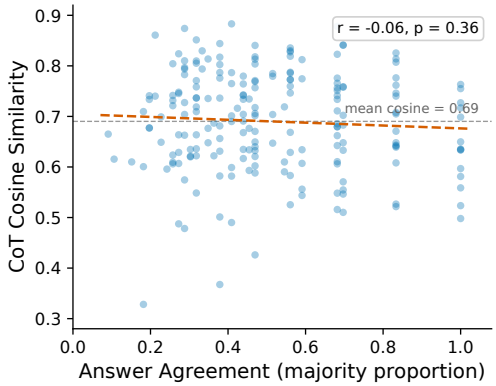


Figure 5: Answer agreement vs. CoT cosine similarity (Mistral-7B, FOLIO, n=204). Flat regression ($r = -0.06$, $p = 0.36$); stronger models show significant positive correlation ($r = +0.24$, Qwen-14B; Table 21).

tive majority does not provide independent signal; it reinforces the majority answer that SC would already select. This confirmation bias, not a lack of discrimination, explains why SICA degenerates to majority voting (Appendix H).

L Hyperparameter Sensitivity

L.1 K Sensitivity

Figure 6 shows SICA Δ as a function of the number of sampled traces K for Mistral-7B on FOLIO ($n = 204$, no fixed seed). The relationship is non-monotonic: $K = 4$ yields $\Delta = -1.47$ pp, $K = 8$ yields $\Delta = -1.96$ pp, $K = 12$ reaches $\Delta = +3.43$ pp, $K = 16$ drops to $\Delta = +1.47$ pp, and $K = 20$ partially recovers to $\Delta = +1.96$ pp. The recovery at $K = 20$ rules out a simple peaked pattern, but $K = 12$ remains the only setting with $\Delta > 2$ pp ($p = 0.092$, marginal).

The apparent Δ peak at $K = 12$ is confounded with SC baseline. SICA absolute accuracy plateaus at 59.31% (121/204) for $K \geq 12$: identical at $K = 12$ and $K = 20$. The Δ variation across these settings is driven primarily by SC fluctuations rather than by SICA improvement. $K = 12$ produces a relatively low SC (55.88%), placing it in the intermediate-SC region; $K = 8$ (SC = 58.33%) and $K = 20$ (SC = 57.35%) have higher baselines, compressing Δ even though SICA accuracy does not change for $K \geq 12$. Additional traces beyond $K = 12$ provide no benefit to SICA absolute performance, consistent with the constraint quality bottleneck. The $K = 12$ Δ captures only 25.0% of the oracle ceiling (+13.73 pp).

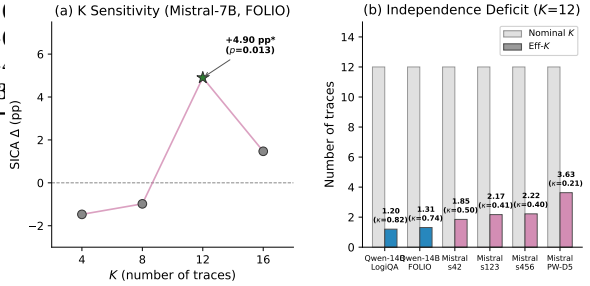


Figure 6: SICA Δ (pp) vs. number of traces K for Mistral-7B on FOLIO ($n = 204$, default seed, $T = 0.7$). Non-monotonic pattern: SICA accuracy plateaus at 59.31% for $K \geq 12$; the apparent Δ peak at $K = 12$ partly reflects a relatively low SC baseline (55.88%) at that point. Error bars: 95% bootstrap CI.

L.2 Temperature Sensitivity

Temperature sensitivity on Mistral-7B FOLIO ($n = 204$, default seed) reveals a sharp isolated peak rather than a smooth trend. Across four temperature settings, only $T = 0.7$ produces a positive effect: $T = 0.3$ yields $\Delta = -1.47$ pp (SC = 58.33%), $T = 0.5$ yields $\Delta = -0.98$ pp (SC = 58.33%), $T = 0.7$ yields $\Delta = +3.43$ pp (SC = 55.88%), and $T = 1.0$ yields $\Delta = 0.00$ pp (SC = 55.39%). All other temperatures yield negative or null effects, and the single positive result at $T = 0.7$ is confounded with SC baseline. Combined with the seed sensitivity (0 of 4 seeds reaching $p < 0.05$ at $T = 0.7$), the positive trend depends on a specific combination of temperature, seed, and baseline accuracy that same-model sampling rarely achieves.

The parallel between K and T sensitivity points to a common underlying variable. In both sweeps, the setting that produces the largest SICA gain also produces a relatively low SC baseline (55.88%), which places it in the intermediate-SC region. The hyperparameters themselves (K, T) are not the causal driver; they modulate SC level, and it is the SC level that determines whether SICA has operating room.

M Entropy Decomposition

We partition problems by trace agreement: high-entropy (majority count ≤ 8 out of $K = 12$) and low-entropy (majority count ≥ 9). Table 15 reports

Table 14: Additional intervention results beyond the nine prompt-level and three method-level strategies reported in the main text. All produce neutral-to-negative results, completing the exhaustive remediation analysis across 27 distinct approaches.

#	Method	Type	Target	Δ (pp)	Key Finding
13	PROVE (code verif.)	Paradigm	Formal proof	-1.9	15.2% compilable; conclusive acc. = 50%
14	CEGIS (iterative)	Multi-round	Consistency	$0.0^{\dagger}$	R1 $+3.43$ pp then oscillates
15	DeepConf (logprob)	Filtering	Confidence	0.0	$\Delta_{\text{logprob}} = 0.006$ (no separation)
16	Cross-model SICA	Cross-model	Independence	-0.49	κ_{cross}=0.42 vs κ_{within}=0.53
17	Problem perturbation	Input diversity	Trace diversity	<0	κ *increased* to 0.70 (anti-effect)
18	MIN-UNSAT	Solver objective	Constraint sel.	-6.5	Mean excluded constraints = 1.3
19	κ-aware SC	Voting rule	Weighted voting	0.0	11/11 conditions $\Delta = 0$
20	Z3 trace filter	Trace filtering	Quality	-0.49	Precision 49.1%, 7 questions affected
21	Bootstrapped gold	Iterative	Self-improvement	$0.0^{\dagger}$	R1 $+2.45$ pp then oscillates
22	Selective abstention	Diagnostic	Confidence	$+4.4^{*}$	AUROC = 0.673, at 80% coverage
23	ASCII perturbation	Input diversity	Format diversity	0.0	κ: 0.39 $\rightarrow$ 0.36 (negligible)
24	Atom pipeline	Paradigm	Decomposition	-10.0	Atomic extraction degrades
25	Premise-only filter	Constraint sel.	Bias isolation	$+2.45$	$p = 0.359$; halves conclusion-bias
26	DPP trace selection	Trace diversity	Determinantal	$+1.47$	$p = 0.629$; K_{sel}=8
27	Z3 strict-mode	Solver config	UNSAT filtering	0.0	McNemar b=1, c=1; 63% UNSAT

All experiments on Mistral-7B FOLIO (n=204) unless noted. † Oscillates without convergence; net effect ≈ 0. * Accuracy gain only at reduced coverage (diagnostic value, not a general improvement). Methods 1–12 are reported in Table 12 and Table 13.

SICA Δ within each partition across eight model-dataset conditions.

Table 15: Entropy decomposition of SICA Δ (pp). High-entropy: majority count ≤ 8; low-entropy: majority count ≥ 9. †Partial data (180/600). Pre-tiebreaking SC values; pattern robust to D064 correction.

Model	Dataset	High-entropy		Low-entropy	
		n	Δ	n	Δ
Mistral-7B	FOLIO	78	$\mathbf{+12.8}$	126	0.0
Mistral-7B (s123)	FOLIO	77	$\mathbf{+9.1}$	127	0.0
Mistral-7B (s456)	FOLIO	75	-1.3	129	0.0
Mistral-7B	PW-D5	267	-4.9	333	-1.2
Qwen2.5-14B	FOLIO	37	-8.2	167	-0.6
Qwen2.5-14B	LogiQA	23	-4.3	177	0.0
Qwen2.5-7B	FOLIO	47	-2.1	157	0.0
LLaMA-3.1-8B†	PW-D5	76	-3.9	104	-1.9

Two patterns emerge. First, low-entropy problems produce $\Delta \approx 0$ pp across all eight conditions: when traces agree, constraints carry no information beyond the majority vote.

Second, the direction of SICA's effect on high-entropy problems depends on the accuracy regime, not on entropy alone. In the intermediate-SC region (Mistral-7B on FOLIO, default-seed and seed-123 runs), high-entropy Δ reaches $+12.8$ and $+9.1$ pp: traces genuinely disagree, and constraints from correct-answer traces provide discriminative signal that resolves the disagreement. In Type A conditions (Qwen2.5-14B on FOLIO, $\Delta = -8.2$ pp), high-entropy problems represent rare deviations from a strong model's consensus; constraints inherit the model's confirmation bias. In Type C (Mistral-7B on ProofWriter, $\Delta = -4.9$ pp), high-entropy problems reflect systematic errors. The mechanism is not entropy per se but the interaction of entropy with model capability.

N Failure Mode Analysis

The regime-dependent pattern in Table 1 reflects three distinct failure modes of self-extracted logical constraints.

Type A: confirmation bias (SC $\geq$ ~60%). Strong models produce consistent reasoning traces, and constraints extracted from those traces mirror the answer distribution. When most constraints support the same candidate, MAX-SAT excludes few clauses ($|\mathcal{C}^{\setminus}| \approx 0$), the penalty term in Eq. 3 vanishes, and the scoring function degenerates to weighted vote counting (Eq. 4). SICA and SC then produce identical predictions. The ablation in Appendix H confirms this empirically: on Qwen2.5-14B (FOLIO + ProofWriter, $n = 804$), MAX-SAT and count-only baselines agree on 803 of 804 predictions.

Type B: non-formalizable domains. LogiQA problems require pragmatic and contextual reasoning that resists clean logical formalization. Qwen2.5-14B on LogiQA achieves an extraction rate of 86.4%, but the resulting constraints have a contradiction rate of only 0.8%: extracted formulas rarely conflict across traces, leaving MAX-SAT with no logical tension to exploit. Mistral-7B on LogiQA sits in the same SC range (50.00%) yet produces $\Delta = +1.50$ pp ($p = 0.581$); Qwen2.5-14B on the same domain yields $\Delta = -1.00$ pp ($p = 0.625$). Neither model benefits, confirming that the right accuracy regime is necessary but not

sufficient when the domain resists formalization.

Type C: error amplification (SC $\leq$ ∼40%). When model accuracy is low, the majority of traces reach incorrect answers. Constraints extracted from wrong traces encode incorrect reasoning as formal rules. Z3 treats these rules as soft clauses with weight proportional to their frequency, so the solver preferentially satisfies the wrong constraints and penalizes traces that happen to reach the correct answer. Mistral-7B on ProofWriter ($\Delta = -2.83$ pp, raw $p = 0.014$; BH adj. $p = 0.112$) demonstrates this effect: SICA converts a weak positive signal in trace diversity into a net negative by formalizing the errors.

O Cross-Model Confirmation Bias

Table 16 reports MaxSAT-weighted confirmation bias metrics across three model families on FOLIO ($n = 204$). Weighted bias ratio and wrong-supporting constraint percentage both decrease with model capability, while SICA's accuracy effect shifts from positive (Mistral, intermediate-SC regime) to negative (LLaMA, high-SC regime). Raw bias ratios (4.97×, 3.85×, 2.88×) are reported in §5.1. LLaMA weighted BR CI: [2.19, 3.28].

Table 16: Cross-model confirmation bias on FOLIO ($n = 204$, $K = 12$). BR (wtd.): MaxSAT-weighted ratio of error-supporting constraints on incorrect vs. correct problems. Wrong %: weighted fraction of constraints supporting the incorrect majority on error problems. Δ: SICA − SC (pp).

Model	BR (wtd.)	Wrong (%)	Δ (pp)
Mistral-7B	4.24×	80.9	+4.90
Qwen2.5-14B	3.69×	78.7	−0.98
LLaMA-3.1-8B	2.66×	72.7	−4.41

Frontier model universality. Table 2 (main text, §5.1) extends confirmation bias analysis to nine models spanning 7B open-source to frontier closed-source. All models exhibit BR > 1 (range: 1.66–4.97×). Extraction context varies: open-source models use self-extraction (the model extracts constraints from its own traces); frontier models use cross-model extraction (Mistral-7B extracts FOL from the target model's traces), as closed-source APIs do not expose internal representations suitable for direct symbolic extraction.

ProcessSim definition. Process similarity is the mean pairwise cosine similarity of TF-IDF trace representations:

$$\text{ProcessSim}(q) = \frac{2}{K(K-1)} \sum_{i<j} \cos(\mathbf{v}_i^q, \mathbf{v}_j^q), \tag{7}$$

where $\mathbf{v}_i^q$ is the TF-IDF vector of trace t_i for question q.

Process-κ gradient. Under cross-model extraction (Mistral-7B extracts FOL from other models' traces), SICA gain decreases monotonically with process-level cosine similarity (Table 17; Spearman $\rho = -1.0$, $n = 5$, one-tailed $p = 0.0083$). Gain requires two conditions: diverse traces (low κ_{proc}) *and* substantial self-bias (high BR). R1-Distill-8B satisfies both ($\kappa_{\text{proc}} = 0.248$, BR $= 2.92$) and gains $+2.45$ pp; o3 has moderate trace diversity ($\kappa_{\text{proc}} = 0.596$) but near-minimal bias (BR $= 1.66$, exhibiting a bimodal error pattern: 82% of error problems have BR $= \infty$ but low finite-BR), yielding $\Delta = 0$.

Table 17: Process-κ gradient under cross-model extraction (Mistral-7B → target model traces, FOLIO). κ_{proc}: process-level cosine similarity. Δ: SICA − SC (pp). Spearman $\rho(\kappa_{\text{proc}} \to \Delta) = -1.0$, $p = 0.0083$. *Interim ($n = 95/204$).

Model	κ_{proc}	Δ (pp)	BR
R1-Distill-8B	0.248	+2.45	2.92
gpt-5.5	0.563	+0.49	3.48
o3	0.596	0.00	1.66
Gemini-2.5-Pro*	0.790	−1.05	2.79
gpt-4.1	0.792	−1.47	2.70

P Independence-Accuracy Scatter Plot

Table 18: Independence-accuracy trade-off (FOLIO, $n = 204$). Top: aggregation methods on Mistral-7B; bottom: SC across three model families. All families produce $\kappa \geq 0.41$, confirming an architecture-independent ceiling. κ: Fleiss' kappa on raw answers.

Model	Aggregation	κ	Eff-K	K	Acc. (%)
Mistral-7B	Debate	0.15	4.44	12	38.24
	Diverse-Prompt	0.40	2.21	12	54.41
	Self-Consistency	0.44	2.02	10	56.86
	ICG	0.47	1.93	10	51.96
LLaMA-3.1-8B	Self-Consistency	0.41	2.17	12	66.18
Qwen2.5-7B	Self-Consistency	0.66	1.46	12	75.98
Qwen2.5-14B	Self-Consistency	0.74	1.32	12	75.98

Trace independence metrics. Table 19 reports the full 15-condition breakdown of Fleiss' κ and Effective-K at $K = 12$, referenced in §3.1.

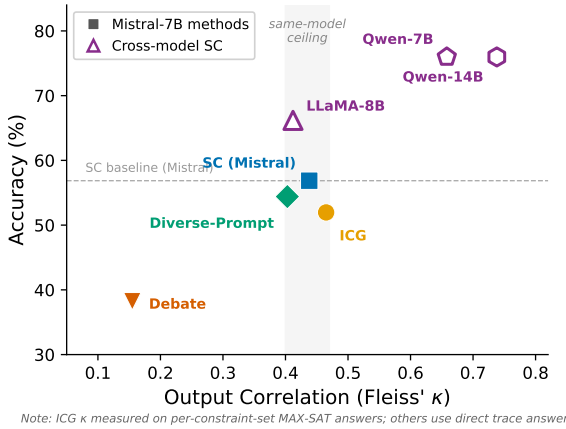


Figure 7: Independence-accuracy trade-off across aggregation methods (Mistral-7B, filled markers) and model families (LLaMA-3.1-8B, Qwen2.5-7B, Qwen2.5-14B; open markers) on FOLIO ($n = 204$). Same-model methods cluster at $\kappa \approx 0.40$–0.47 (gray band), while cross-model SC shows stronger correlations ($\kappa = 0.41$–0.74) at higher accuracy. Debate breaks correlation ($\kappa = 0.15$) at catastrophic accuracy cost. Dashed line: Mistral-7B SC baseline.

Q Cross-Model Kappa Analysis

Table 20 compares trace independence across three model families on the same benchmarks. On FOLIO ($n = 204$), all four models produce $\kappa \geq 0.41$: LLaMA-3.1-8B ($\kappa = 0.41$), Mistral-7B ($\kappa = 0.44$), Qwen2.5-7B ($\kappa = 0.66$), and Qwen2.5-14B ($\kappa = 0.74$). The bottleneck is architecture-independent—no family escapes the correlation floor. Within the Qwen family, κ increases from 0.66 (7B) to 0.74 (14B), but Mistral-7B (7B) and LLaMA-3.1-8B (8B) differ by $\Delta\kappa = 0.03$ despite belonging to different families, suggesting that cross-family variation ($\Delta\kappa$ up to 0.22 at similar scale) exceeds within-family scale effects ($\Delta\kappa = 0.08$). Disentangling scale from architecture requires controlled same-family experiments across $\geq$3 scales.

R Cross-Model Process-Level Independence

Table 21 extends the process-level independence analysis (§5.1) to four model families on FOLIO ($n = 204$). Mean pairwise cosine similarity of TF-IDF trace representations exceeds answer-level κ in every family, but the gap is model-strength dependent: it narrows from 0.31 (LLaMA) and 0.23 (Mistral) to 0.02 (Qwen-14B). For weaker models ($r \approx 0$), answer diversity does not reflect reasoning diversity—process-convergent but answer-divergent trace pairs constitute 9.5% of all pairs on Mistral-7B versus 2.5–3.2% on other families. For stronger models ($r \approx 0.23$, $p \leq 0.001$), process and answer agreement align, and answer-level κ already captures the bottleneck. Process-κ thus provides a crucial additional diagnostic specifically for weaker models, where answer-level metrics underestimate reasoning homogeneity.

Table 19: Trace independence metrics ($K = 12$, sorted by κ descending). κ: Fleiss' Kappa. K_{eff}: Effective-K (Eq. 1). SC: Self-Consistency baseline. Δ: SICA − SC (pp). * $p < 0.05$ (uncorrected; BH adj. $p = 0.112$, m=16); positive Δ values do not reach significance. multi-T = $T \in \{0.3, 0.7, 1.2\}$, 4 traces each.

Model	Condition	κ	K_{eff}	SC (%)	Δ (pp)
Qwen3 (think)	FOLIO	0.91	1.09	86.27	+0.49
Qwen2.5-14B	LogiQA	0.82	1.20	79.00	−1.00
Qwen3	FOLIO	0.77	1.26	83.33	+1.47
Qwen2.5-14B	FOLIO	0.74	1.30	75.98	−0.49
Qwen2.5-7B	FOLIO	0.66	1.46	75.98	+0.49
Mistral-7B (T=0.3)	FOLIO	0.53	1.77	58.33	−1.47
Mistral-7B	LogiQA	0.51	1.81	50.00	+1.50
Mistral-7B (s456)	FOLIO	0.51	1.82	55.88	0.00
Mistral-7B	FOLIO	0.50	1.84	55.88	+3.43
Mistral-7B (s123)	FOLIO	0.49	1.88	54.41	+2.94
Mistral-7B (s789)	FOLIO	0.45	2.03	59.31	−2.45
Mistral-7B (T=0.5)	FOLIO	0.52	1.80	58.33	−0.98
Mistral-7B (multi-T)	FOLIO	0.48	1.91	57.84	0.00
Mistral-7B (T=1.0)	FOLIO	0.48	1.91	55.39	0.00
Mistral-7B	PW-D5	0.21	3.62	39.33	**−2.83***

Table 20: Cross-model trace independence ($K = 12$, SC aggregation, FOLIO + ProofWriter). Higher κ indicates stronger correlation and lower effective sample size.

Model	Dataset	κ	K_{eff}	SC (%)
Qwen2.5-14B	FOLIO	0.74	1.32	75.98
Qwen2.5-7B	FOLIO	0.66	1.46	75.98
Mistral-7B	FOLIO	0.44	2.02	56.86
LLaMA-3.1-8B	FOLIO	0.41	2.17	66.18
Qwen2.5-14B	PW-D5	0.62	1.54	—
Mistral-7B	PW-D5	0.21	3.62	39.33

Representation robustness. Replacing TF-IDF with SBERT embeddings (all-MiniLM-L6-v2) yields ProcessSim = 0.91 (vs. TF-IDF 0.74), widening the gap over answer-level $\kappa = 0.55$. The process homogeneity finding is robust to representation choice.

S Constraint Quality Across Methods

Table 22 compares constraint quality metrics across the three method-level interventions on Mistral-7B FOLIO ($n = 204$). Standard SICA extracts from reasoning traces and achieves the highest parse rate (97.1%) but also the highest redundancy (42.9%): nearly half of all constraints are syntactic duplicates across the 12 traces, quantifying the co-sourcing

Table 21: Cross-model process-level independence on FOLIO ($n = 204$, $K = 12$). Mean Cosine: mean pairwise TF-IDF cosine similarity of reasoning traces (Eq. 7). Answer κ: per-question answer agreement. Gap: Cosine $-$ κ. r: Pearson correlation between ProcessSim and answer agreement. Conv.: fraction of process-convergent but answer-divergent trace pairs. ** $p < 0.01$; *** $p < 0.001$.

Model	Mean Cos.	Ans. κ	Gap	r	p	Conv. (%)
Mistral-7B	0.690	0.460	0.230	-0.064	0.362	9.5
LLaMA-3.1-8B	0.759	0.447	0.312	-0.001	0.988	3.2
Qwen2.5-7B	0.790	0.658	0.132	$+0.226$	0.001**	2.8
Qwen2.5-14B	0.766	0.744	0.022	$+0.244$	<0.001***	2.5

effect of same-model generation. ICG generates constraints directly from the problem statement without trace conditioning. Its lower parse rate (74.6%) reflects the difficulty of standalone formalization at the 7B scale, though successfully parsed constraints compile at a higher rate (99.4% vs. 96.5%). ICG produces fewer unique constraints per question (24.9 vs. 39.3) with lower redundancy (32.3%), consistent with the absence of trace-level repetition. Contrastive scoring generates pro and con constraint sets for each candidate answer. The format yields far fewer unique constraints per question (9.2) with zero redundancy (each pro/con pair is structurally distinct), but 62.7% of constraints are trivially satisfied by Z3—they impose no logical tension and cannot discriminate between candidates. Only 51.0% of Contrastive constraints are discriminative, compared to 81.4% for standard SICA and 77.0% for ICG.

Table 22: Constraint quality metrics across method-level interventions (Mistral-7B, FOLIO, $n = 204$). Parse: fraction of extraction attempts yielding valid JSON. Z3 Compile: fraction compiling to Z3 formulas. Unique/q: mean unique constraints per question. Redundancy: fraction of constraints duplicated across traces. Disc.: fraction discriminative (different Z3 outcome per candidate). Triv. Sat.: fraction trivially satisfied (no logical tension).

Method	Parse	Z3	Uniq/q	Redund.	Disc.	Triv. Sat.
Standard SICA	97.1%	96.5%	39.3	42.9%	81.4%	6.2%
ICG	74.6%	99.4%	24.9	32.3%	77.0%	6.2%
Contrastive	74.0%	94.4%	9.2	0.0%	51.0%	62.7%

T Error Categorization of Method-Level Interventions

We categorize discordant predictions (where the method and SC disagree) for Debate and Contrastive on FOLIO ($n = 204$) to understand why these interventions degrade accuracy.

Debate discordant analysis. Debate produces 60 correct→wrong flips and 22 wrong→correct flips on FOLIO. Answer-type breakdown reveals that False-labeled questions are most vulnerable: 82.8% of False questions that SC answers correctly are flipped by Debate, compared to 63.9% for True and 25.5% for Unknown. Counterintuitively, failures concentrate not where debate breaks down but where it succeeds most thoroughly: discordant questions average 2.93 effective debate rounds (proposer-critic exchanges producing substantive updates) versus 2.52 for concordant ones. More extensive same-model debate produces more confident but incorrect constraints, consistent with the independence limitation—the critic draws on the same model distribution as the proposer, reinforcing shared biases rather than correcting them.

Contrastive discordant analysis. Contrastive produces 34 discordant questions on FOLIO. All 34 produced non-degenerate scores (both pro and con constraint sets non-empty), compared to 20.1% degenerate scores in the full sample. This indicates that scoring failures on discordant questions stem from systematic pro-bias rather than missing constraints. Among discordant questions, 97.1% contain negation words in the problem statement versus 76.5% in the full sample, suggesting that negation-bearing problems are particularly susceptible to pro-constraint imbalance. The average pro/con score ratio is 1.0 for discordant questions versus 0.68 for the full sample: the method generates balanced-looking scores that mask directional errors.

Cross-method overlap. Of the 79 unique questions on which either method disagrees with SC, only 15 overlap between Debate and Contrastive (Jaccard index $= 0.19$). All 15 shared discordant questions contain negation words. The low overlap confirms that the two methods attack different vulnerability types—Debate through extended same-model argumentation that amplifies confident errors, Contrastive through pro-bias in constraint generation—while sharing a common root cause in correlated same-model voting: both methods rely on the same model's output distribution and cannot introduce genuinely independent signal.

U TOST Equivalence Tests

We apply two one-sided tests (TOST) (Lakens, 2017) with equivalence margin ± 2 pp to all 16

non-degenerate SICA–SC comparisons, classifying each condition as *equivalent* (TOST significant), *genuinely different* (McNemar $p < 0.05$ and TOST not significant), or *underpowered* (neither test significant). Four conditions are equivalent, confirming that SICA and SC produce statistically indistinguishable accuracy. Two conditions show genuine differences, both negative: Mistral-7B ProofWriter (-2.83 pp, $p = 0.014$) and Qwen2.5-14B ProofWriter (-2.50 pp, $p = 0.011$). The remaining ten conditions are underpowered, reflecting the limited statistical power at n=200–204 for detecting $|\Delta| \leq 2$ pp effects.

Margin sensitivity. Table 23 shows that the classification is robust across margins: regardless of the chosen margin, no condition shows a significant positive SICA effect.

Table 23: TOST classification sensitivity across equivalence margins (16 non-degenerate conditions).

Margin	Equivalent	Different	Underpowered
±1 pp	0	2	14
±2 pp	4	2	10
±3 pp	10	0	6

Relative-margin TOST. The absolute ±2 pp margin is more stringent at high baselines (±2.4% relative at SC = 85%) than low ones (±5.1% at SC = 39%). A relative ±5% of SC margin addresses this asymmetry: 8 conditions are equivalent, 2 genuinely different (both negative), and 6 underpowered. The additional four equivalent conditions are all high-SC ($\geq 75\%$), where the wider relative band captures practical indistinguishability. No condition shows a significant positive effect under either margin.

Table 24: Conditional Correction Ratio across eight verifier–generator conditions. CCR > 0: cross-arch NLI on ProofWriter. CCR < 0: same-model SICA or weak capability gap.

Generator–Verifier	Dataset	CCR	Δ (pp)
Mistral × BART-NLI	PW	+1.86	+5.17
Mistral × DeBERTa-NLI	PW	+1.67	+6.67
Mistral × RoBERTa-NLI	PW	+1.42	+4.17
Qwen-14B × DeBERTa-NLI	PW	−0.21	+0.33
Mistral × DeBERTa-NLI	FOLIO	−0.47	+2.94
Mistral × SICA	FOLIO	−2.93	+3.43
Qwen-14B × SICA	PW	−4.21	−2.50
Mistral × SICA	PW	−4.61	−2.83

Conditional Correction Ratio data.

Disagreement-Routed Cascade details. DRC invokes NLI only when SC consensus falls below threshold τ, avoiding unnecessary verifier calls on high-agreement problems. Gains concentrate on weak generators (SC $< 40\%$), consistent with the capability-gap hypothesis (§6). Regime-dependent bottleneck analysis appears in Appendix W.

Table 25: DRC on Mistral-7B (K=12, DeBERTa-large-mnli, w=3). τ=0.7 recovers 83–90% of full-combo gain at ∼35% NLI cost across both datasets.

Dataset	Method	Acc.	NLI Calls	Cost	Recovery
PW-D5	SC only	39.33%	0/600	0%	—
	DRC τ=0.7	**45.33%**	**207/600**	**34.5%**	**90.0%**
	Full combo	46.00%	600/600	100%	100%
FOLIO	SC only	55.88%	0/204	0%	—
	DRC τ=0.7	**58.33%**	**71/204**	**34.8%**	**83.3%**
	Full combo	58.82%	204/204	100%	100%

V Multi-Verifier NLI Results

Table 26: Cross-architecture NLI combo on ProofWriter depth-5 (n=600, Mistral-7B traces, SC = 39.33%). Three NLI verifiers × three weight settings. $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ (McNemar).

Verifier	Standalone	w	Δ (pp)	p
DeBERTa-MNLI	55.00%	1	+0.83	n.s.
		3	+6.67	<0.001***
		5	+11.00	<0.001***
RoBERTa-MNLI	37.67%	1	−0.17	n.s.
		3	+4.17	0.003**
		5	+4.33	0.037*
BART-MNLI	46.17%	1	0.00	n.s.
		3	+5.17	<0.001***
		5	+6.33	0.003**

All verifiers fine-tuned on MultiNLI. Unknown recall <3.5% across all verifiers.

Three-verifier BH correction. Table 27 applies Benjamini–Hochberg correction across all 15 Mistral-7B verifier–weight conditions (m=15, α=0.05; distinct from the m=12 cross-generator analysis in §6). Ten of 15 conditions remain significant after correction, confirming that multi-verifier NLI gains are robust to multiple testing.

Table 27: BH correction across 3 verifiers × 5 weights on Mistral-7B ProofWriter (n=600, m=15). ✓: $p_{\text{BH}} < 0.05$.

Verifier	w=1	w=2	w=3	w=4	w=5
DeBERTa-MNLI		✓	✓	✓	✓
RoBERTa-MNLI			✓	✓	✓
BART-MNLI			✓	✓	✓

10/15 significant at α=0.05 after BH correction. All w=1 conditions and RoBERTa w=2 are non-significant.

W Capability Gap Data Points

Table 3 reports the controlled cross-generator comparison with DeBERTa-large-mnli on ProofWriter

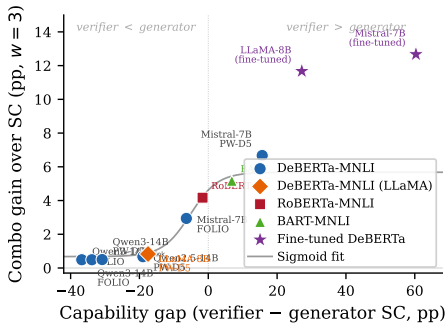


Figure 8: Combo gain vs. capability gap (verifier standalone − generator SC) at w=3 across three NLI architectures. Gain decreases monotonically (Spearman $\rho = 1.0$, exact permutation $p = 0.003$, n=6; four of six conditions share the DeBERTa verifier—see Limitations); significant gains occur only for generators below the verifier's standalone accuracy. Stars: domain-adapted DeBERTa (99.67% standalone, Table 29).

(n=600). We extend this analysis by pairing Mistral-7B with two additional NLI architectures: BART-large-mnli (46.17% standalone, combo +5.17 pp, $p = 0.0002$) and RoBERTa-large-mnli (37.67% standalone, combo +4.17 pp, $p = 0.003$). All three NLI architectures produce significant gains on Mistral-7B at $w \geq 3$.

Across all six ProofWriter verifier-generator pairs at w=3 (three verifiers × Mistral-7B plus three negative-gap generators × DeBERTa), combo gain decreases monotonically with the capability gap (Spearman $\rho = 1.0$, exact permutation $p = 0.003$, $n = 6$; however, four of six conditions share the DeBERTa verifier, which may reduce effective degrees of freedom—see Limitations; Figure 8). Significant gains are observed only for Mistral-7B (SC 39.33%, below the verifier); all generators with SC $\geq$ 72% show no significant improvement. After BH correction (m=12), two of twelve conditions survive: Mistral-7B w=3 (adj. $p = 6.78 \times 10^{-5}$) and w=5 (adj. $p = 7.42 \times 10^{-6}$).

Robustness check. Including all nine conditions (three verifiers × three weight levels on Mistral-7B) yields Spearman $\rho = 0.836$ (exact permutation $p = 0.007$, $n = 9$), confirming that the monotonic trend is robust to the choice of weight parameter.

Full NLI combo with BH correction.

Domain-adapted verifier results.

Generator×Extractor 2×2 Matrix. Table 30 reports the full 2×2 generator×extractor matrix on FOLIO (n=204). The generator dominates accuracy (row difference ∼25 pp vs. column difference ∼3 pp), contrasting with ProofWriter where the extractor dominates (Δ range ±40–50 pp). Cross-extractor McNemar tests: Mistral-7B gen, Δ = +2.94 pp, $p = 0.143$ (n.s.); Qwen3-14B gen, Δ = +2.94 pp, $p = 0.021$ (statistically significant but practically small: 6/204 problems changed). The identical net Δ is coincidental—the two generators have different discordant-pair distributions that happen to yield the same net advantage. BR > 1 in all eight cells across both datasets (range 1.94–2.70), confirming SCB universality under cross-model extraction.

Table 28: Full cross-generator NLI combo results on ProofWriter depth-5 (n=600) with Benjamini–Hochberg correction (m=12, α=0.05). All conditions use DeBERTa-large-mnli verifier (55.00% standalone). Two of twelve conditions survive correction (Mistral-7B w=3 and w=5). *** $p_{BH} < 0.001$.

Generator	w	Combo (%)	Δ (pp)	p_{raw}	p_{BH}	
Mistral-7B	1	40.17	+0.83	.359	.720	
SC 39.33%, gap +15.67	3	46.00	+6.67	1.13×10^{-5}	6.78×10^{-5}	***
	5	50.33	+11.00	6.18×10^{-7}	7.42×10^{-6}	***
LLaMA-3.1-8B	1	73.33	+0.83	.359	.720	
SC 72.50%, gap −17.50	3	73.33	+0.83	.665	.887	
	5	72.00	−0.50	.871	1.0	
Qwen2.5-14B	1	74.17	+0.17	1.000	1.0	
SC 74.00%, gap −19.00	3	74.67	+0.67	.503	.755	
	5	75.17	+1.17	.360	.720	
Qwen3-14B	1	86.00	+0.17	1.0	1.0	
SC 85.83%, gap −30.83	3	86.33	+0.50	.4531	.755	
	5	87.00	+1.17	.2100	.720	

Table 29: Domain-adapted NLI combo on ProofWriter depth-5 (n=600). DeBERTa-large fine-tuned on ProofWriter OWA (99.67% standalone). All $p < 0.0001$. Complementarity: ratio of NLI-correct∩SC-incorrect to SC-correct∩NLI-incorrect.

Generator	w	Combo (%)	Δ (pp)	Compl. ratio
Mistral-7B	1	42.17	+2.83	
SC 39.33%	3	52.00	+12.67	364:2
	5	65.50	+26.17	
LLaMA-3.1-8B	1	76.00	+3.50	
SC 72.50%	3	84.17	+11.67	164:1
	5	91.67	+19.17	

Table 30: FOLIO 2×2 generator×extractor matrix (n=204). SC: Self-Consistency baseline. κ: answer-level Fleiss' kappa. BR: bias ratio.

Generator	Extractor	SC (%)	SICA (%)	Δ (pp)	BR
Mistral-7B	Mistral	55.88	56.86	+0.98	2.70
κ=0.46	Qwen3	55.88	59.80	+3.92	1.94
Qwen3-14B	Qwen3	83.33	84.80	+1.47	2.35
κ=0.79	Mistral	83.33	81.86	−1.47	2.66

X Single-Trace and Aggregation Statistics

Table 31: Single-trace accuracy (mean ± std over K=12 traces) and SC@12 majority vote on FOLIO (n=204). Aggregation gain = SC@12 − single-trace mean. SC@12 uses alphabetical tiebreaking (consistent with Table 1).

Model	Single (%)	Std	SC@12 (%)	Gain (pp)
Mistral-7B	46.41	2.80	55.88	+9.47
Qwen3-14B	79.37	1.98	84.80	+5.43
Qwen3-14B (think)	85.42	0.99	86.27	+0.86

Inter-trace standard deviation shrinks 2.8× from Mistral-7B to Qwen3-14B thinking mode, reflecting the diversity compression that underlies the aggregation ceiling (§6).

Table 32: Effect of extraction temperature T_{ext} on SICA accuracy (Mistral-7B, FOLIO n=204). † Uses a different trace set (SC = 54.41%); all other conditions share the same traces (SC = 55.88%).

T_{ext}	SICA (%)	Δ vs. SC (pp)	Avg. #Constr.
0.1	57.35	+1.47	69.3
0.3†	59.31	+3.43	68.8
0.5	56.37	+0.49	68.2
0.7	53.43	−2.45	67.3
1.0	55.39	−0.49	65.2

Extraction temperature sensitivity. Controlling for the trace set (excluding $T_{\text{ext}} = 0.3$, which uses a different seed), all conditions yield $\Delta \leq +1.47$ pp (n.s.), indicating that extraction temperature has no significant effect on SICA accuracy. This supports the structural nature of symbolic confirmation bias: the bottleneck is what gets extracted, not the temperature at which extraction occurs.

Table 33: Multi-seed replication results. Each row is an independent seed with fresh traces (K=12, T=0.7). Mean ± SD computed across seeds. Mistral-7B shows directionally unstable Δ (+0.16 ± 2.70 pp); all Qwen2.5-14B seeds show Δ ≤ 0.

Seed	SC (%)	SICA (%)	Δ (pp)
Mistral-7B × FOLIO (n=204)[a]			
123	54.41	57.35	+2.94
456	55.88	55.88	0.00
789	59.31	56.86	−2.45
Mean ± SD	56.53 ± 2.51	56.70 ± 0.75	+0.16 ± 2.70
Qwen2.5-14B × FOLIO (n=204)			
42	77.50	75.49	−2.01
123	74.51	73.04	−1.47
456	74.02	74.02	0.00
789	74.02	72.55	−1.47
Mean ± SD	75.01 ± 1.67	73.78 ± 1.30	−1.24 ± 0.86

[a]Original baseline (exp033) used no fixed seed and is excluded from seed statistics.

Multi-seed replication.

X.1 Sub-population Analysis

On FOLIO-204 with the corrected SC baseline (alphabetical tiebreaking), SICA improves over SC by +3.4 pp overall (55.9% → 59.3%), though the aggregate McNemar test is marginally non-significant (p=.092). The gain concentrates in the mid-constraint tertile (+9.4 pp, p=.031), where sufficient symbolic evidence exists for MaxSAT to meaningfully re-rank answers; in the low- and high-constraint groups the methods perform comparably. On ProofWriter-600, SICA shows a small uniform degradation consistent with the main results, attenuating in the high-constraint tertile (Δ=−0.5 pp, p=1.0). We note that with six subgroup tests, one significant result at α=0.05 is expected by chance; the mid-constraint finding should be interpreted as exploratory.

Table 34: SICA vs. SC accuracy by gold-label type (Mistral-7B). Corrected SC baseline uses alphabetical tiebreaking. Δ pp = SICA − SC. p-values from exact McNemar test (two-sided).

Dataset	Label	n	SC	SICA	Δ pp	p
FOLIO-204	True	72	56.9	62.5	+5.6	.125
	False	63	42.9	42.9	±0.0	1.0
	Unknown	69	66.7	71.0	+4.3	.250
	Overall	204	55.9	59.3	+3.4	.092
PW-600	True	200	8.5	6.0	−2.5	.063
	False	200	19.5	14.5	−5.0	.076
	Unknown	200	90.0	89.0	−1.0	.774
	Overall	600	39.3	36.5	−2.8	.014*

Table 35: SICA vs. SC accuracy by extracted constraint count tertile (Mistral-7B). Corrected SC baseline (alphabetical tiebreaking). Constraint counts are unique deduplicated constraints per problem.

Dataset	Tertile	n	$\bar{c}$	SC	SICA	Δ pp	p
FOLIO-204	Low [0–32]	74	21.7	62.2	63.5	+1.4	1.0
	Mid [33–47]	64	39.7	50.0	59.4	+9.4	.031*
	High [48–83]	66	58.7	54.5	54.5	±0.0	1.0
PW-600	Low [0–39]	212	28.8	43.9	39.6	−4.3	.078
	Mid [40–52]	196	45.8	40.8	37.2	−3.6	.143
	High [53–99]	192	63.7	32.8	32.3	−0.5	1.0

Individual intervention significance. Of the 27 training-free remediation strategies (Figure 4), the two largest positive effects are Cross-model Qwen3→Mistral (+0.98 pp) and Generational (+0.49 pp); neither approaches statistical significance ($p > 0.5$ for both). The largest negative effects are LLM-Debate (−18.6 pp, $p < 0.001$) and Contrastive Scoring (−9.3 pp, $p < 0.001$). After BH correction across all 27 comparisons, only these two negative results remain significant. No strategy produces a significant positive effect under any correction procedure.

Y Extended Discussion

SCB versus CoT unfaithfulness. SCB is distinct from CoT unfaithfulness (Turpin et al., 2024) and CoT-level confirmation bias (Wan et al., 2025; Jhaveri et al., 2026). CoT unfaithfulness concerns whether stated reasoning reflects actual computation; SCB concerns whether *self-extracted formal constraints* mirror the model's answer distribution. Even if traces were perfectly faithful, self-extraction would still inherit the distributional bias—the constraint $P(\text{answer_A})$ reflects vote share, not logical validity. The shuffle control (§5.1) confirms this: randomizing constraint–answer associations preserves BR, showing the bias is in the trace-count distribution, not in extraction-specific errors. However, if traces are unfaithful, the mirage may be worse than our measurements suggest, because constraints would reflect not the model's reasoning but its post-hoc rationalization.

Implications for neuro-symbolic practitioners. Our results do not imply that neuro-symbolic methods are inherently flawed—they imply that *self-extracted* constraints cannot provide independent verification. The key distinction is *when* formalization occurs relative to answer generation. Specification-generation approaches (e.g., SatLM (Ye et al., 2024), Logic-LM (Pan et al., 2023)) formalize the problem *before* generating candidate answers, avoiding the extraction bottleneck; SatLM's success confirms that the formal verification mechanism is sound when constraints are independently derived. Trace-extraction (this work) operates post-hoc on reasoning traces that already contain candidates, inheriting the distributional prior that generates SCB. When post-hoc verification is unavoidable, practitioners should employ external verifiers from a different architecture family (§6) and use the diagnostic framework in §7 to assess feasibility before deployment. The SCB diagnostic toolkit (below) provides detailed computation methods and threshold recommendations.

Test-time scaling implications. Vote diversity decreases with model capability: K_{eff} drops from 3.62 (Mistral-7B on ProofWriter) to 1.09 (Qwen3-14B thinking mode). This creates a fundamental tension for o1/R1-style scaling (Guo et al., 2025): longer reasoning chains produce more coherent but less diverse traces, reducing the marginal value of additional samples. Our Process-κ analysis (§5.1) reveals that this homogenization operates at the reasoning-process level, not just answer level—traces that disagree on final answers still share 69% of their reasoning structure. Cross-architecture ensembles offer a path forward, but the capability-gap condition (§6) implies that the verifier must be genuinely complementary, not just different.

SCB diagnostic toolkit. We recommend four zero-cost diagnostics, computable from sampled traces alone, to predict whether symbolic self-verification will help or hurt before deployment. *Bias Ratio* (BR; §5.1): for each problem, compute the fraction of constraints supporting the majority answer, divided by majority vote share. BR > 1 indicates constraints are biased toward the majority; BR $\gg 2$ (as in all nine models tested; Table 2) signals that extraction amplifies rather than checks the model's prior. Report as mean $\pm$ bootstrap 95% CI. *Effective-K* (K_{eff}; Eq. 1): compute Fleiss' κ across K traces, then $K_{\text{eff}} = 1 + (K-1)(1-\kappa)$. Threshold: $K_{\text{eff}} \leq 1.30$ never improves in our experiments—aggregation collapses to majority voting. $K_{\text{eff}} > 1.8$ is a necessary (not sufficient) condition for gain. *Degenerate Consensus* (C_0; §3.1): the fraction of problems where all K traces agree. $C_0 > 95\%$ renders aggregation moot; $C_0 < 5\%$ means the model is maximally uncertain and constraints tend to amplify noise. *ProcessSim* (Eq. 7): mean pairwise cosine similarity of TF-IDF trace vectors. ProcessSim > 0.6 indicates homogeneous reasoning; combined with BR > 2, this predicts SCB will dominate. A practitioner should compute all four on a held-out sample ($n \geq 50$). If $K_{\text{eff}} \leq 1.30$ or $C_0 > 95\%$ or (BR > 2 and ProcessSim > 0.6), symbolic self-verification is unlikely to help.

Future directions: trained extractors and beyond. Our oracle controls reveal that perfect extraction would yield $+29.90$ pp over SC on Mistral-7B FOLIO—self-extraction captures only $\sim$12% of this ceiling. This gap motivates several future directions. *Trained constraint extractors*: fine-tuning a small model (e.g., encoder-only) on human-annotated FOL from logic benchmarks could produce genuinely discriminative constraints that break the distributional prior. The key design constraint is structural independence—the extractor should not share training data or architecture with the generator, mirroring the cross-architecture NLI escape (§6). *Multi-model ensembles*: rather than single cross-architecture pairs, combining M structurally independent verifiers could further re-

duce correlated errors. The capability-gap condition (§6) predicts that each verifier must exceed the generator on the target task to contribute positively; empirical verification across heterogeneous verifier pools is needed. *External knowledge grounding*: connecting to knowledge bases (Wikidata, domain ontologies) could inject verification signal absent from any model's parameters. This avoids SCB entirely by grounding constraints in external facts rather than self-generated traces. *SCB-aware aggregation*: current MAX-SAT aggregation treats all constraints equally. Weighting constraints by estimated independence (e.g., inverse ProcessSim with other traces' constraints) or by discriminative power (e.g., cross-validation on held-out traces) could partially mitigate SCB without requiring external verifiers.

SCB as an evaluation framework. The SCB metrics—BR, K_{eff}, ProcessSim, and CCR—can serve as a general evaluation framework for self-improvement methods beyond SICA. Any method that extracts structured representations from a model's own outputs (self-verification (Huang et al., 2024), self-refinement (Madaan et al., 2023), chain-of-verification (Dhuliawala et al., 2023), self-consistency (Wang et al., 2023)) faces the distributional prior that generates SCB. We propose that future work on self-improvement methods report: (1) BR on error cases (does the method's verification signal correlate with the model's majority answer, or does it provide independent information?); (2) K_{eff} or an analogous diversity metric (does the method generate genuinely diverse candidates, or does it amplify a single mode?); (3) CCR (does the method preferentially correct errors, or does it preferentially confirm them?). A method with BR < 1, high K_{eff}, and CCR > 0 demonstrates genuine self-improvement; BR > 1 with CCR < 0 indicates that the method suffers from SCB regardless of aggregate accuracy gains, which may reflect favorable base rates rather than corrective signal.

Additional scope considerations. Beyond the limitations discussed in the main text, several additional constraints bound our conclusions. Seven of our eleven model-domain conditions have $n \leq 60$, providing insufficient statistical power for confirmatory claims; power analysis shows that the minimum detectable effect exceeds 5 pp for these conditions (Table 6), so they serve as regime characterization rather than definitive tests. Our extraction prompts use a single zero-shot template per domain; alternative prompting strategies (few-shot exemplars, chain-of-thought extraction, or iterative refinement) might yield higher-quality constraints, though the oracle gap suggests the bottleneck is structural rather than prompt-specific. The constraint quality analysis (91.8% discriminative rate, 39.2% source-trace minority weight) covers Qwen2.5-14B on FOLIO + ProofWriter ($N = 804$); other model-domain combinations may exhibit different characteristics. Constraint extraction uses $T_{\text{ext}} = 0.3$ while trace sampling uses $T = 0.7$; alternative extraction temperatures might yield different results. Although we prompt for first-order logic formalizations, 99.0% of extracted constraints are propositional at the Z3 level, with only 1.0% using true quantifiers (∀/∃); whether models capable of genuine quantified reasoning would yield higher discriminative rates is untested. Deduplication uses syntactic equivalence (global average dedup ratio: 46.0%), so semantically equivalent constraints with different surface forms receive separate weights; Z3-based semantic deduplication removes an additional ∼10% of constraints and yields a small but significant improvement over no deduplication ($p = 0.013$), suggesting room for better dedup strategies. The GSM8K pilot ($n = 30$) illustrates the constraint-absent regime and should not be interpreted as confirmatory evidence.